So, today, we're going to talk about **Long Short-Term Memory** (**LSTM**), which is a type of **Recurrent Neural Network** that is designed to avoid the **exploding/ vanishing** gradient problem.



…**Long Short-Term Memory** uses two separate paths to make predictions about tomorrow.

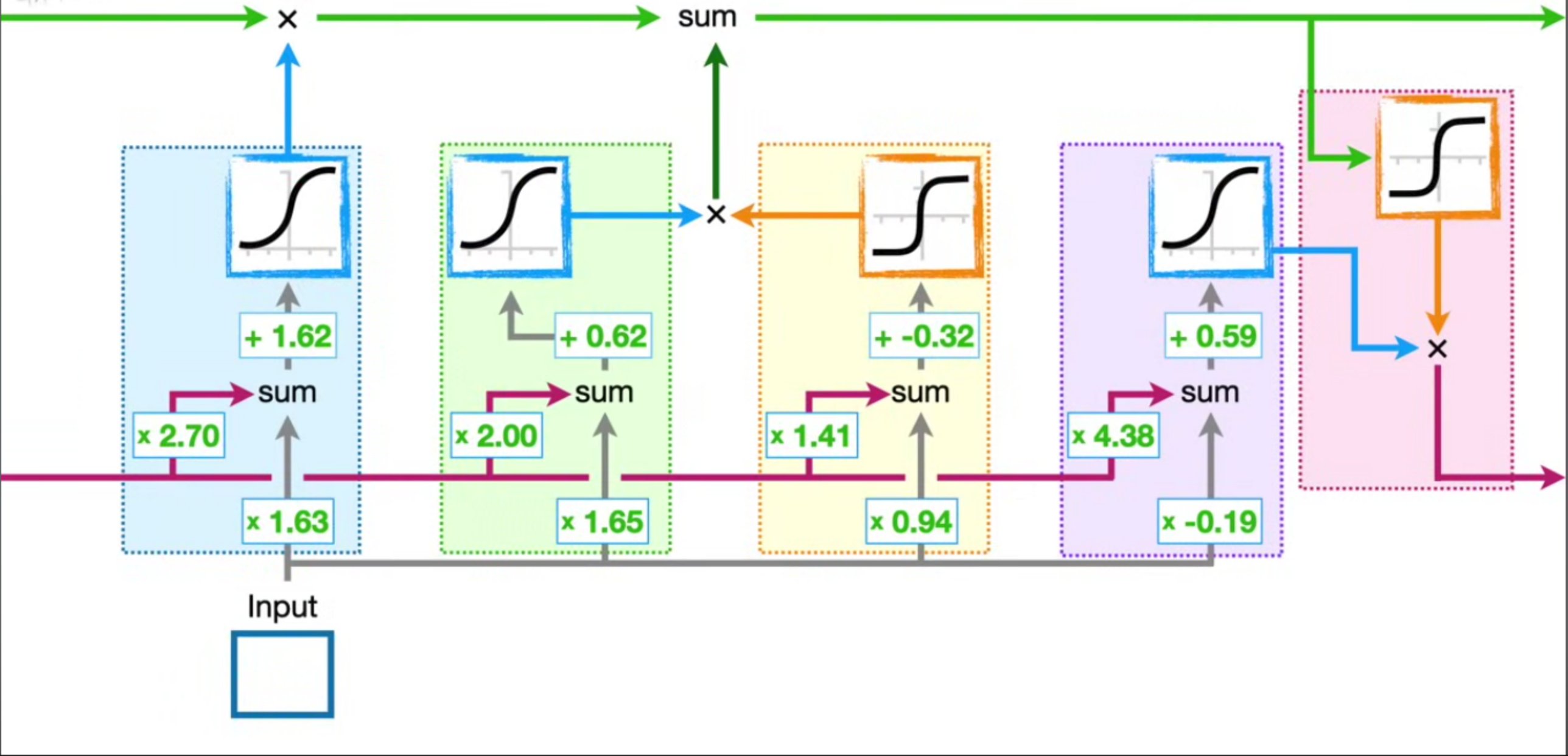


One path is for **Long-Term Memories**…

…and one is for **Short-Term Memories**.

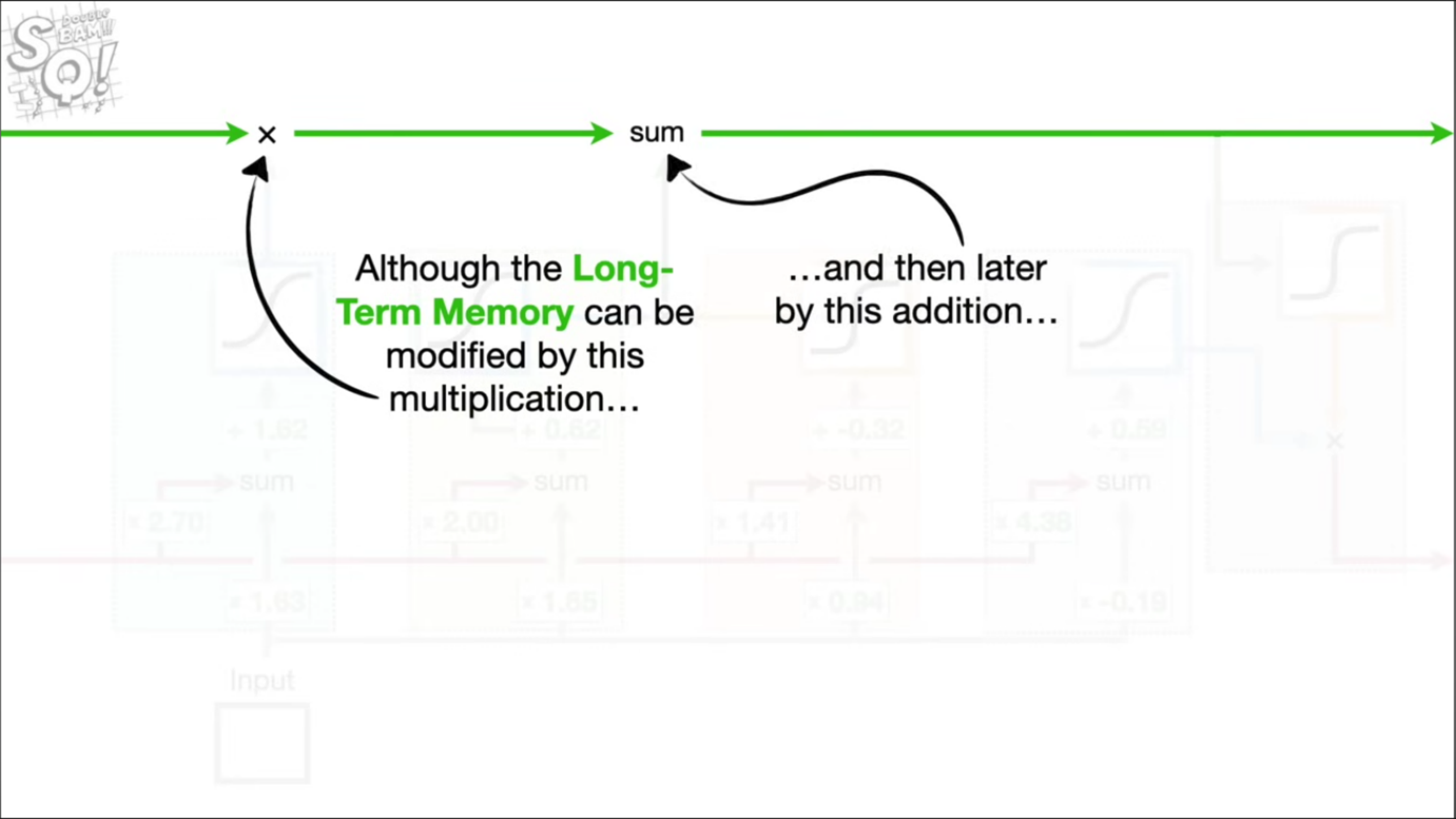


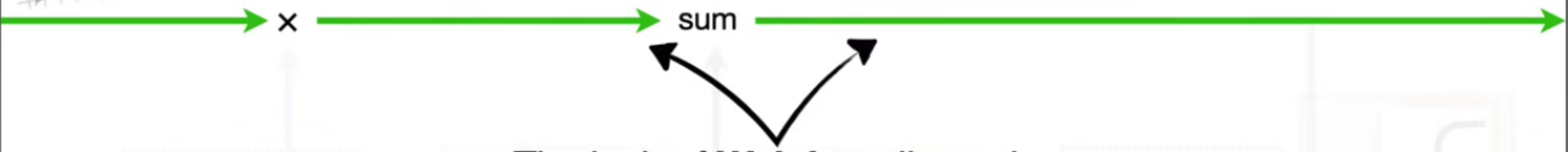
…let's talk about how the **Long Short-Term Memory** unit works.



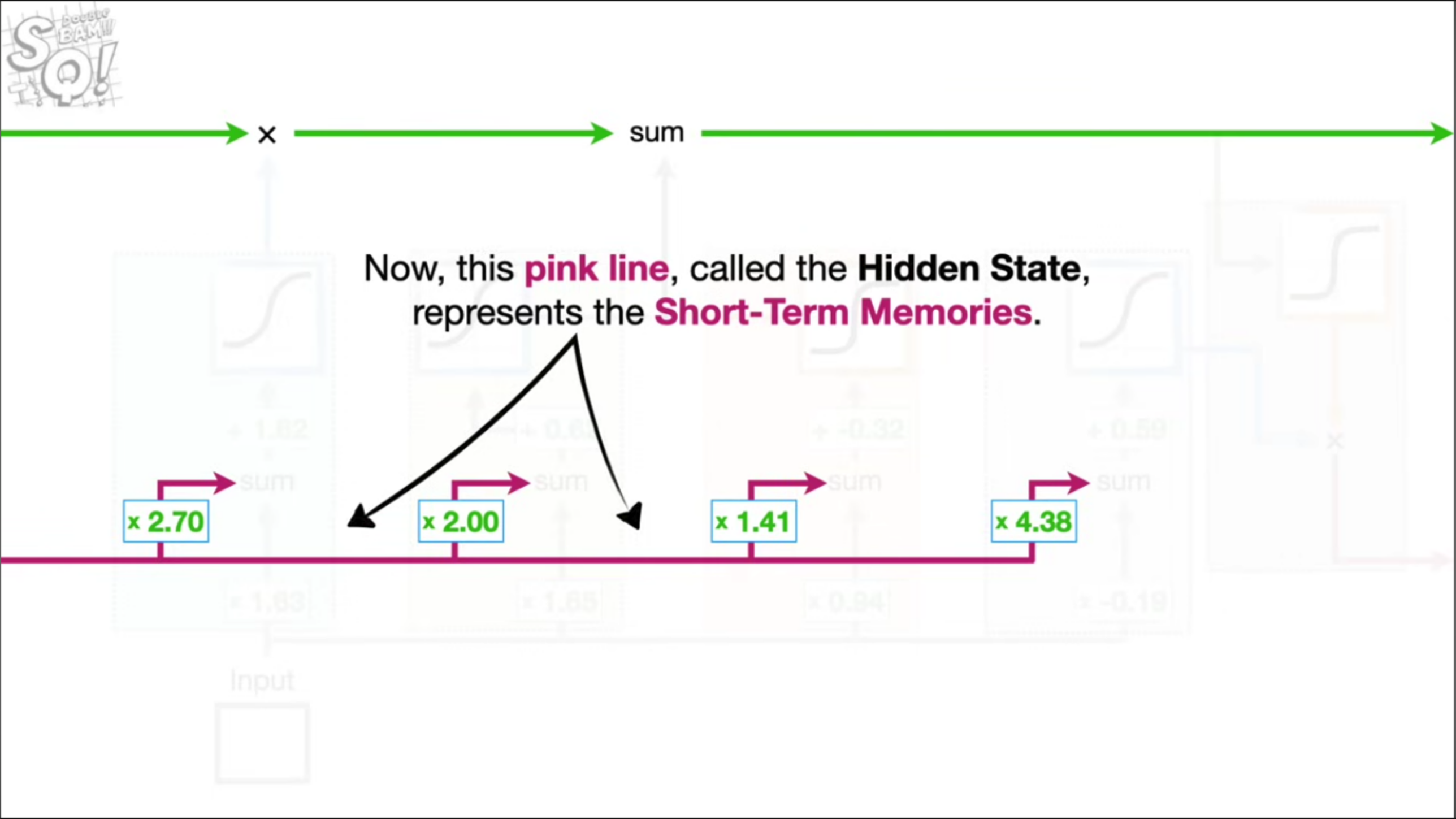


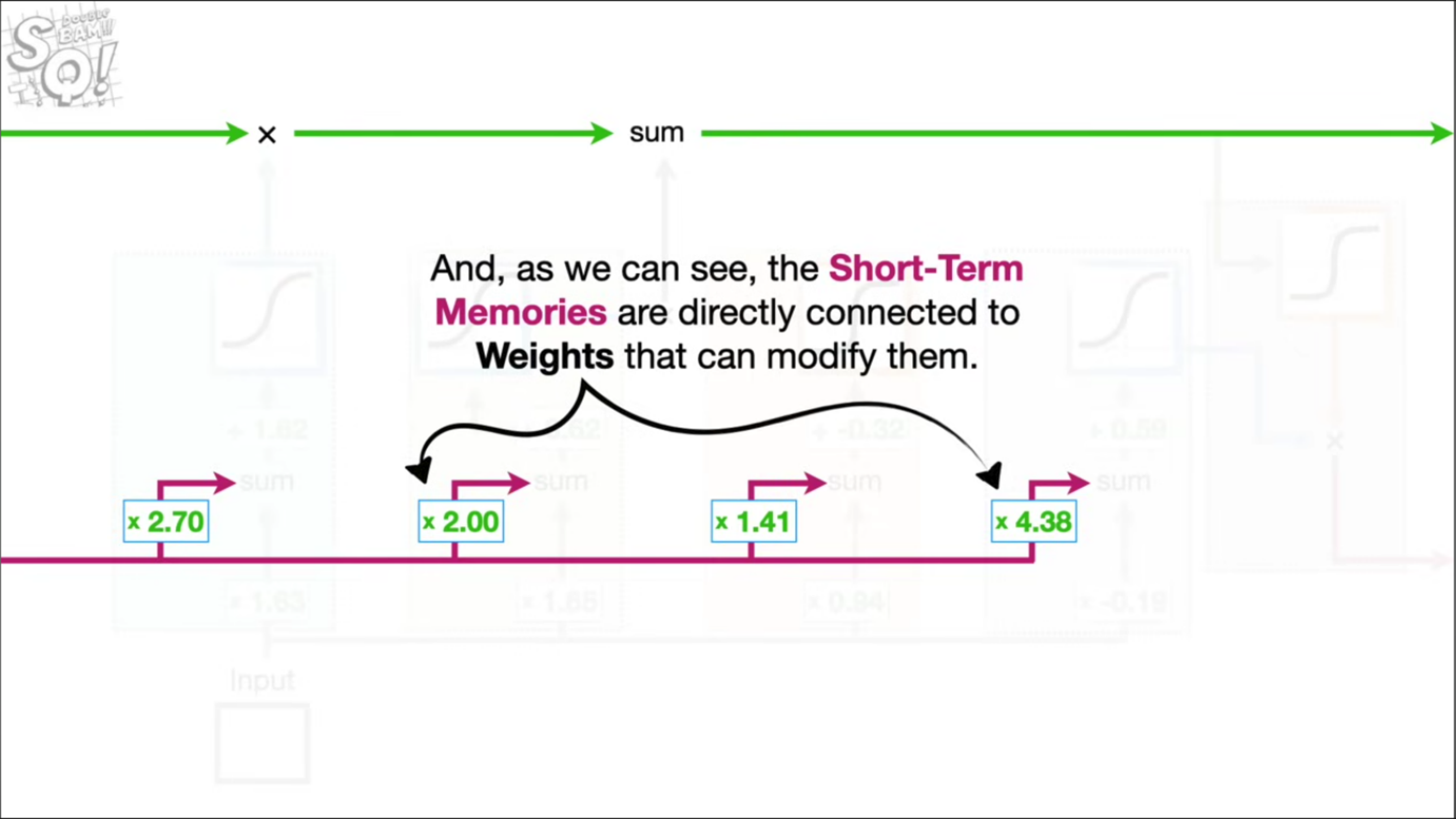
First, this **green line** that runs all the way across the top of the unit is called the **Cell State** and represents the **Long-Term Memory**.

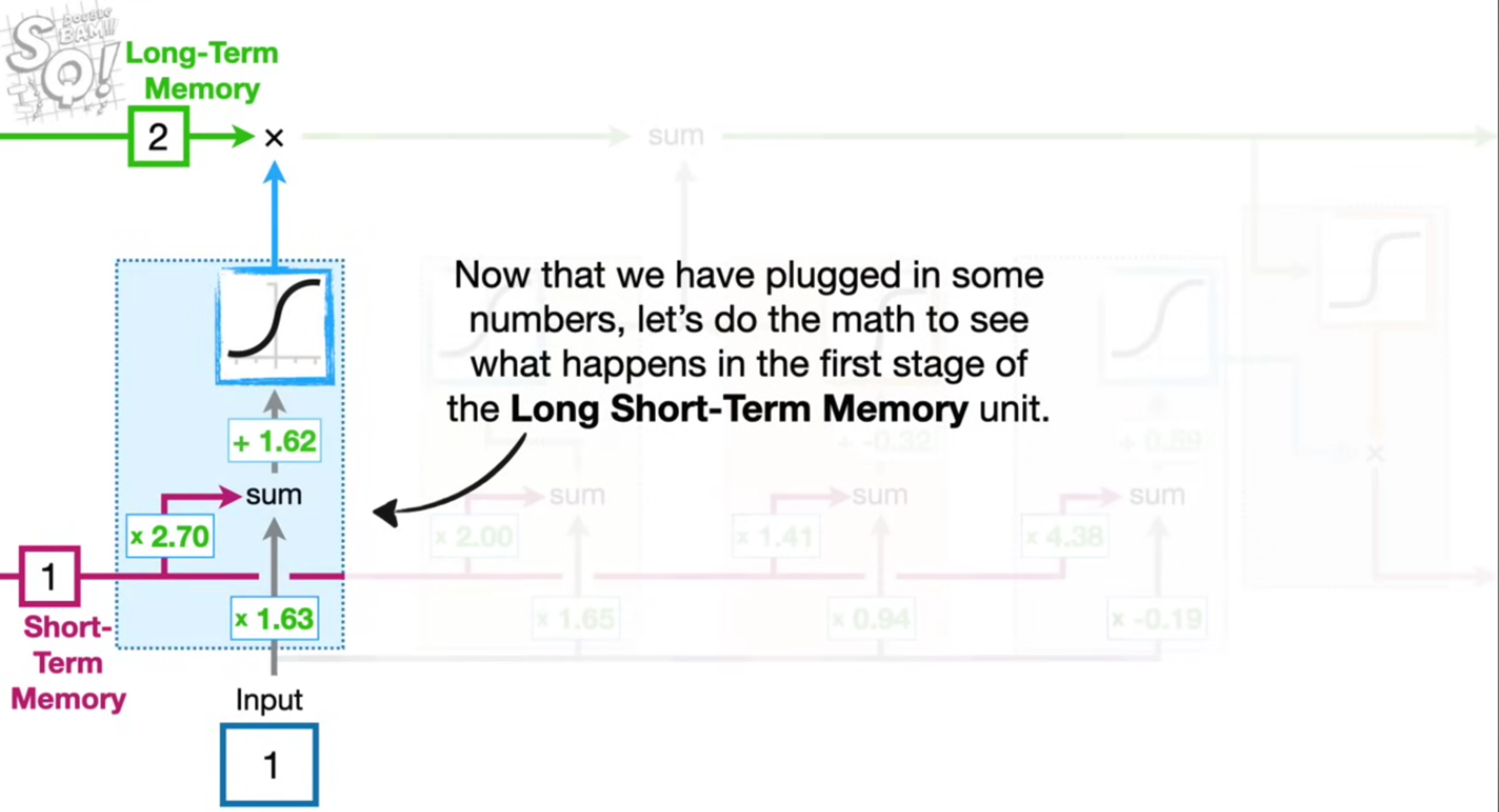

x
sum
Although the Long-Term Memory can be modified by this multiplication…
…and then later by this addition…

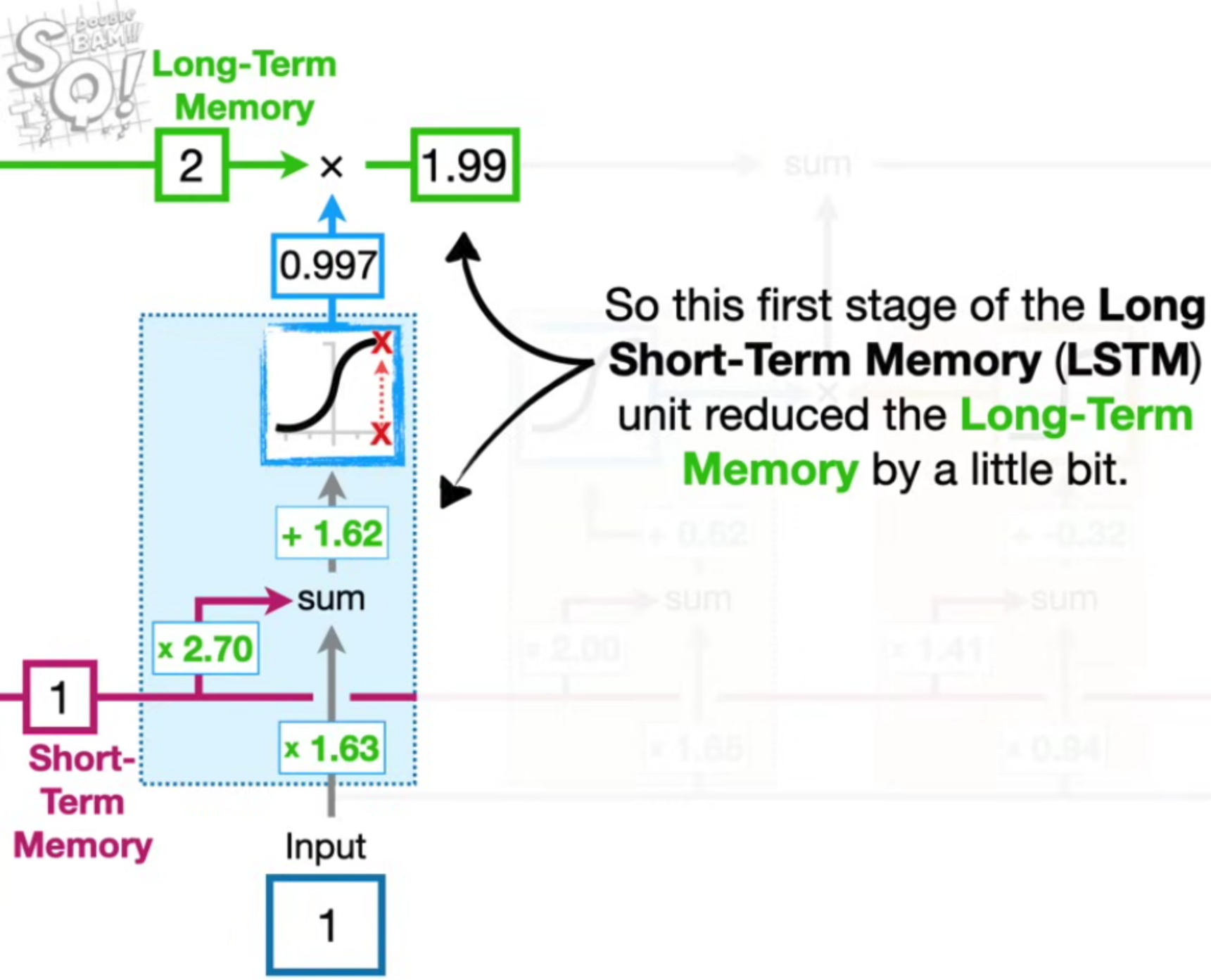


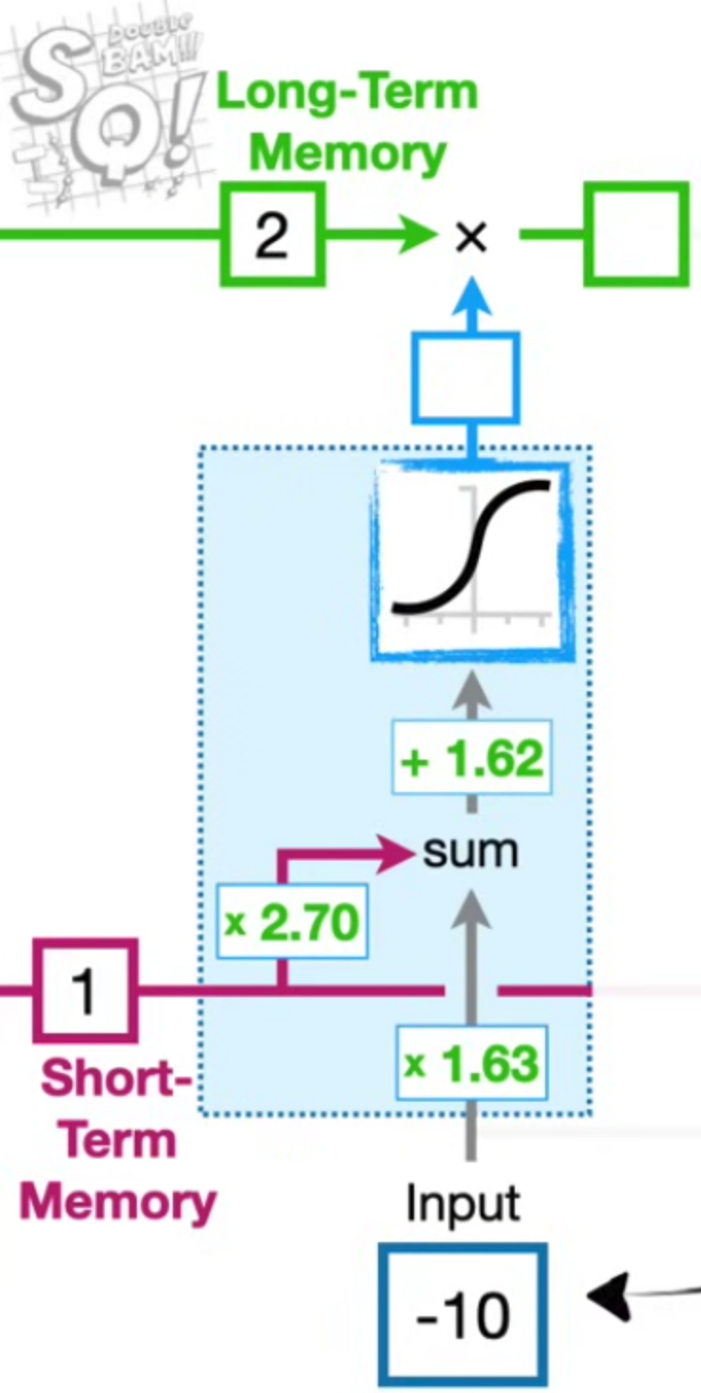
The lack of **Weights** allows the **Long-Term Memories** to flow through a series of unrolled units without causing the gradient to **explode** or **vanish**.


x
sum
Now, this pink line, called the Hidden State,
represents the Short-Term Memories.
x 2.70
x 2.00
x 1.41
x 4.38

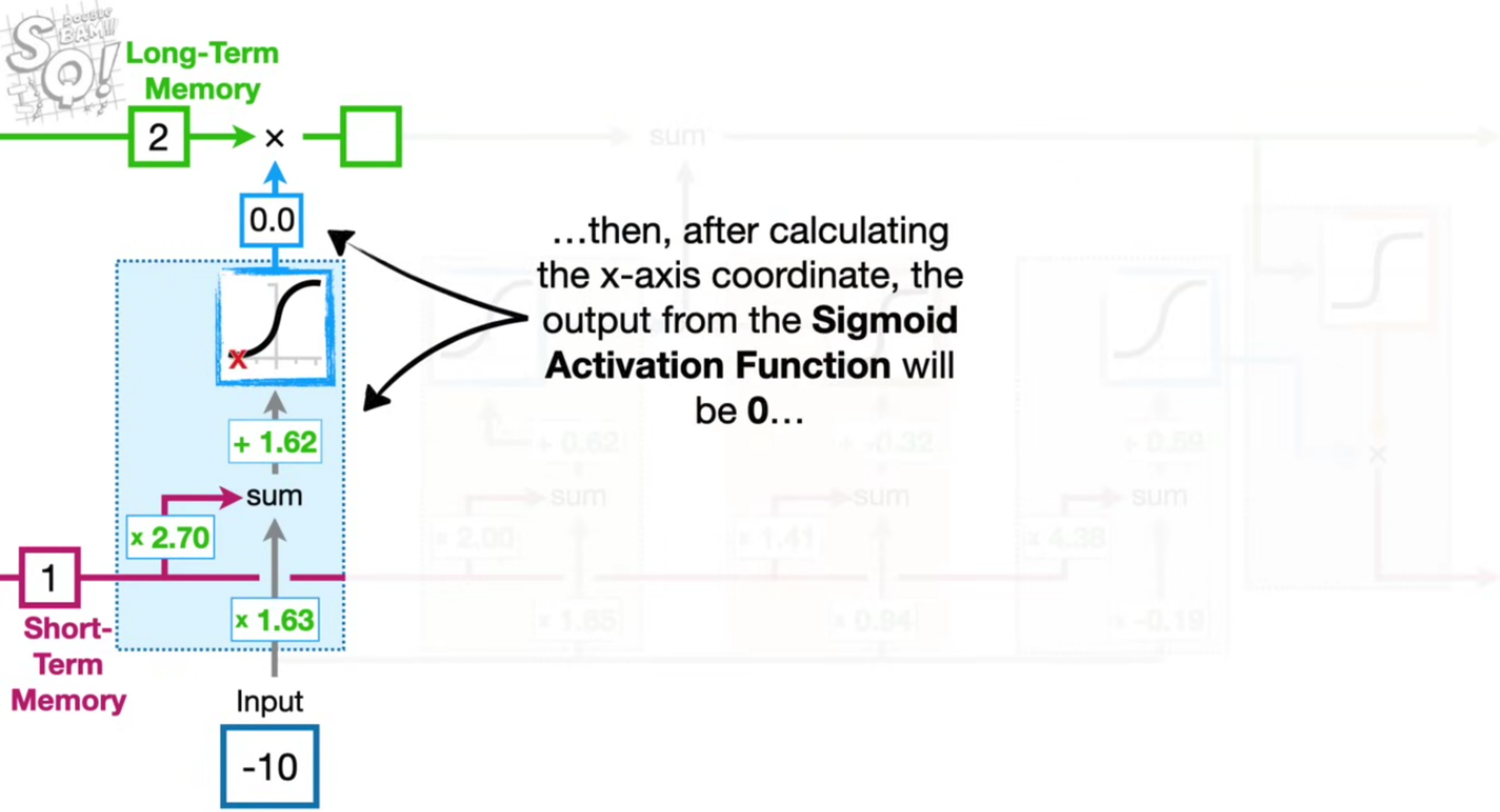

×
sum
And, as we can see, the Short-Term Memories are directly connected to Weights that can modify them.
x 2.70
x 2.00
x 1.41
x 4.38

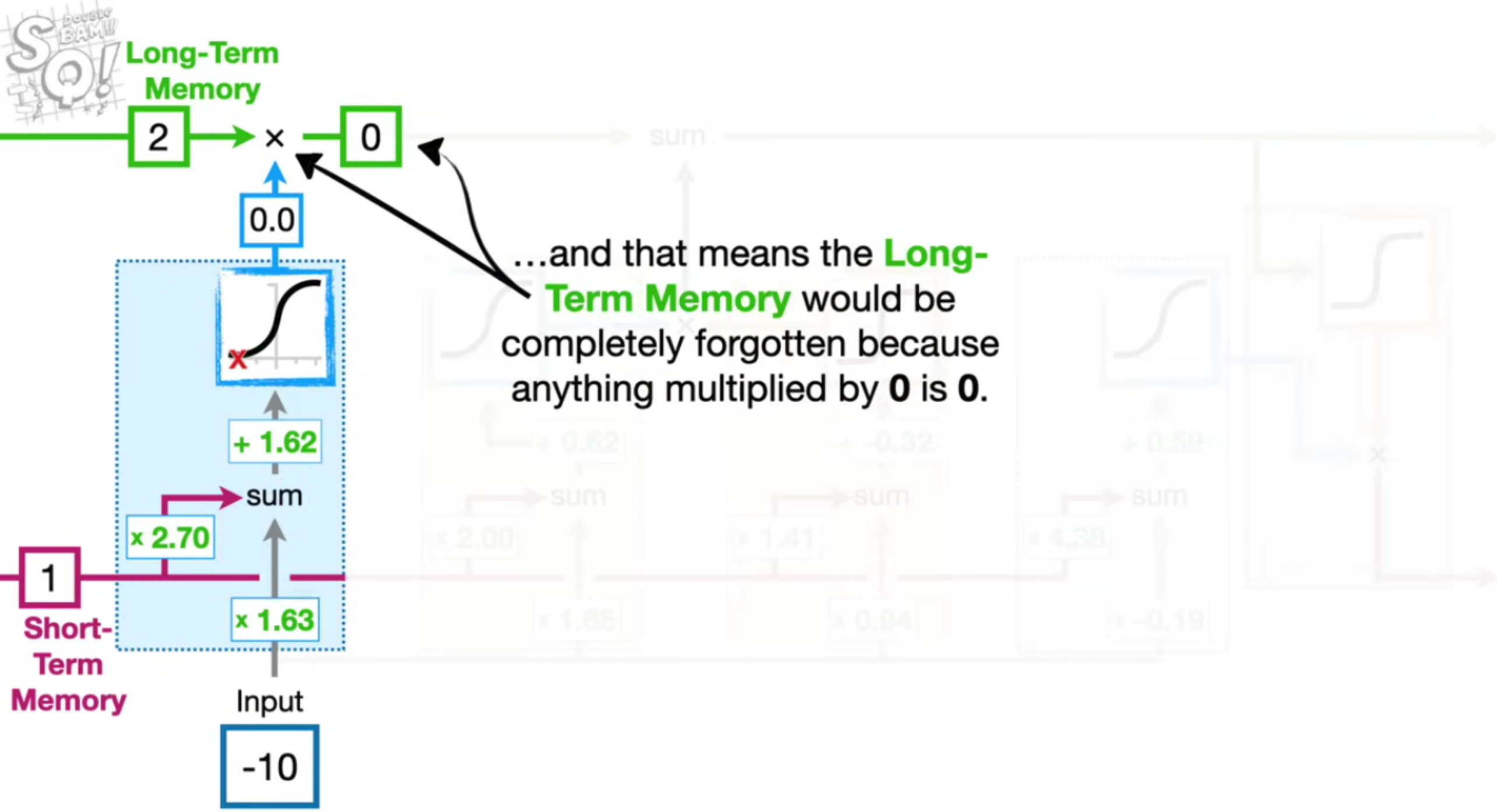

Long-Term Memory
2
×
sum
+ 1.62
sum
x 2.70
1
x 1.63
Short-Term Memory
Input
1
Now that we have plugged in some numbers, let's do the math to see what happens in the first stage of the **Long Short-Term Memory** unit.

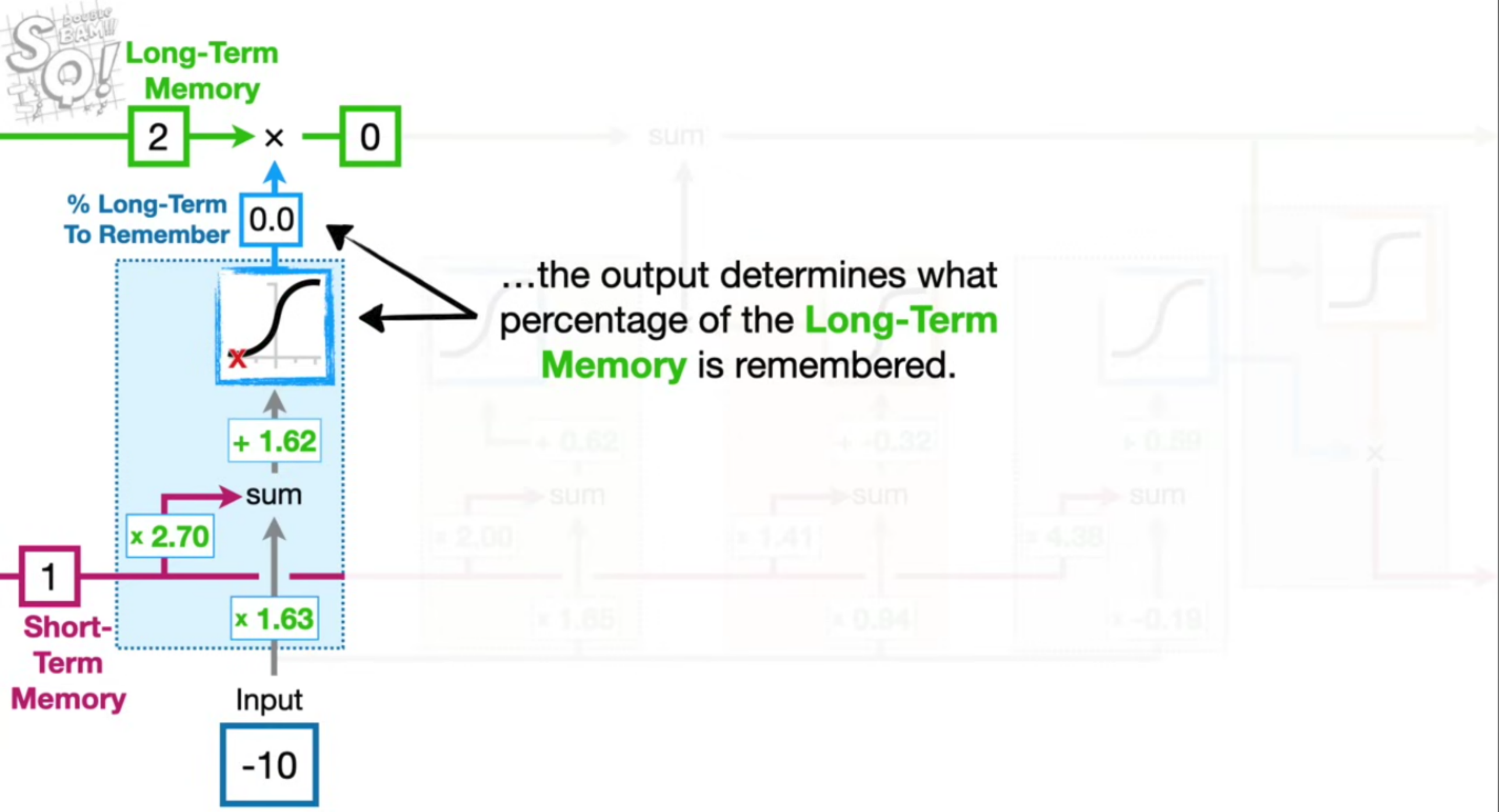

Long-Term Memory
2
×
1.99
0.997
So this first stage of the Long Short-Term Memory (LSTM) unit reduced the Long-Term Memory by a little bit.
+ 1.62
sum
x 2.70
1
x 1.63
Short-Term Memory
Input
1

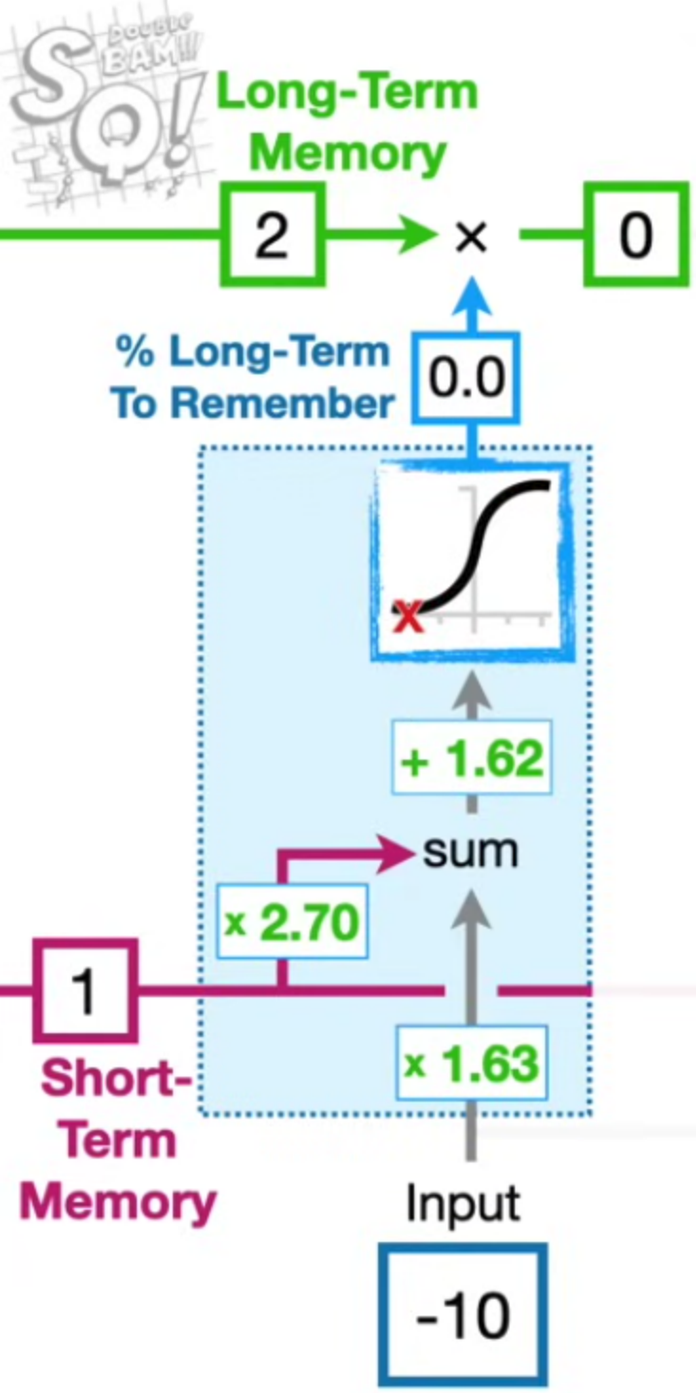


In contrast, if the **Input** to the **LSTM** was a relatively large negative number, like **-10**…

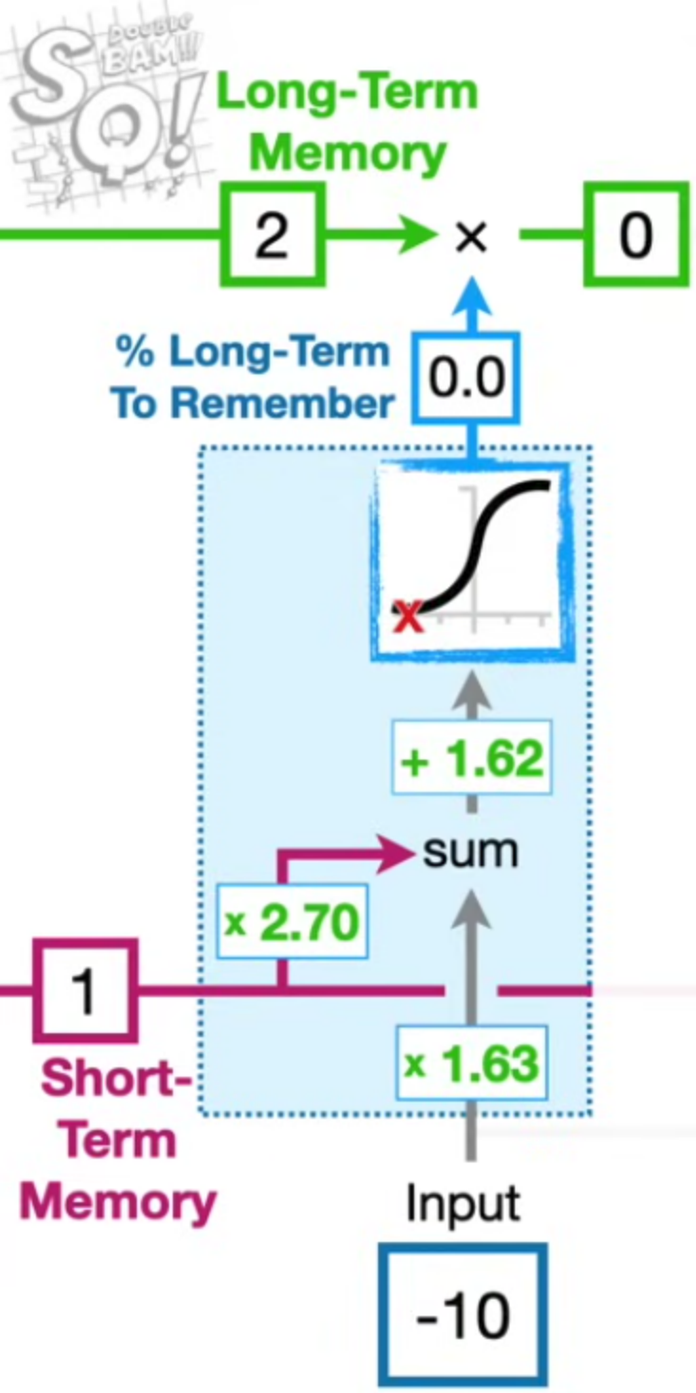

Long-Term Memory
2
×
0.0
…then, after calculating the x-axis coordinate, the output from the **Sigmoid Activation Function** will be **0**…
+ 1.62
sum
x 2.70
1
x 1.63
Short-Term Memory
Input
-10


Long-Term Memory
2
×
0
0.0
…and that means the Long-Term Memory would be completely forgotten because anything multiplied by 0 is 0.
+ 1.62
sum
x 2.70
1
x 1.63
Short-Term Memory
Input
-10


Long-Term Memory
2
×
0
% Long-Term To Remember
0.0
…the output determines what percentage of the Long-Term Memory is remembered.
+ 1.62
sum
x 2.70
1
x 1.63
Short-Term Memory
Input
-10



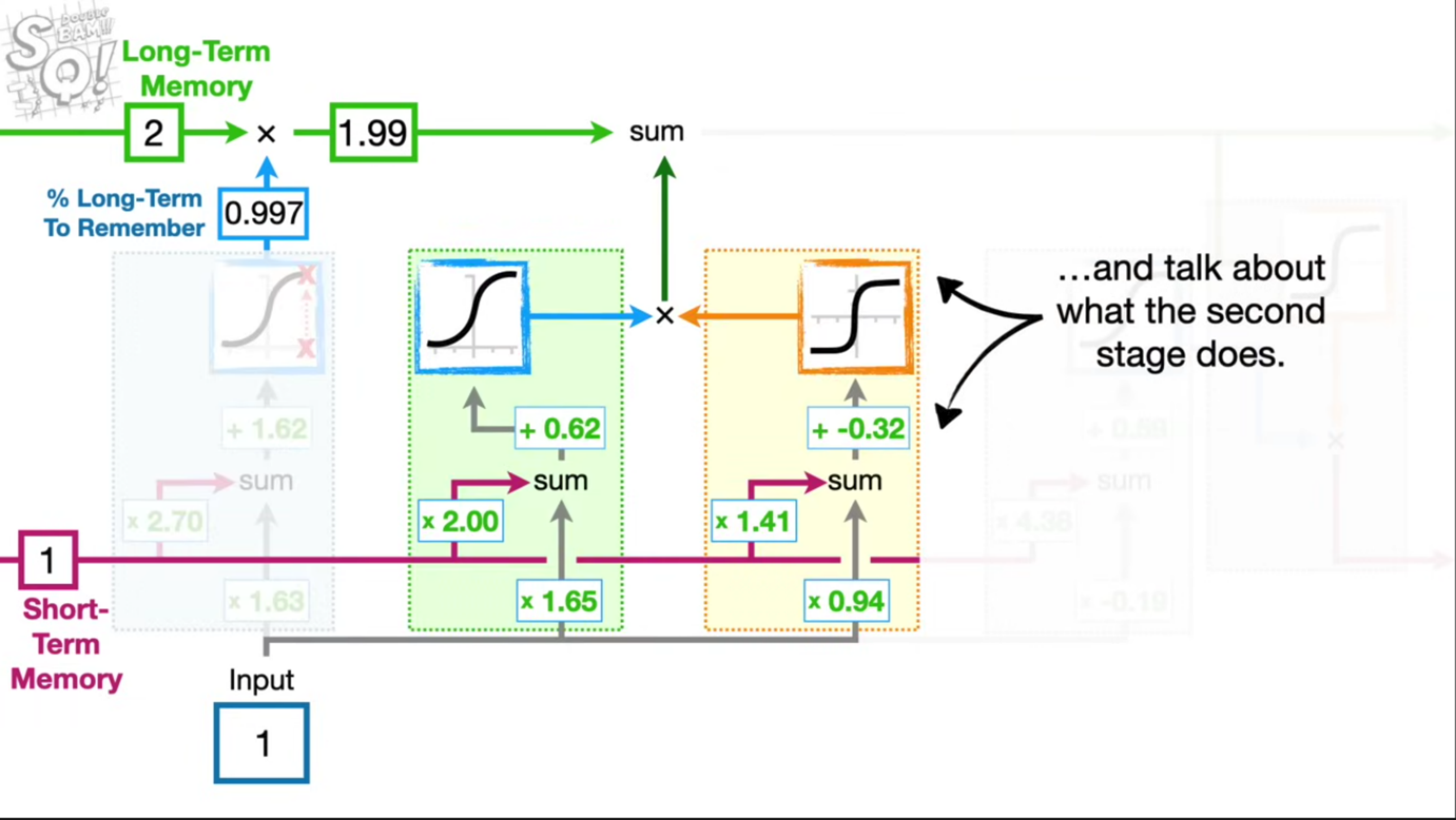
To summarize, the first stage in a **Long Short-Term Memory** unit determines what percentage of the **Long-Term Memory** is remembered.

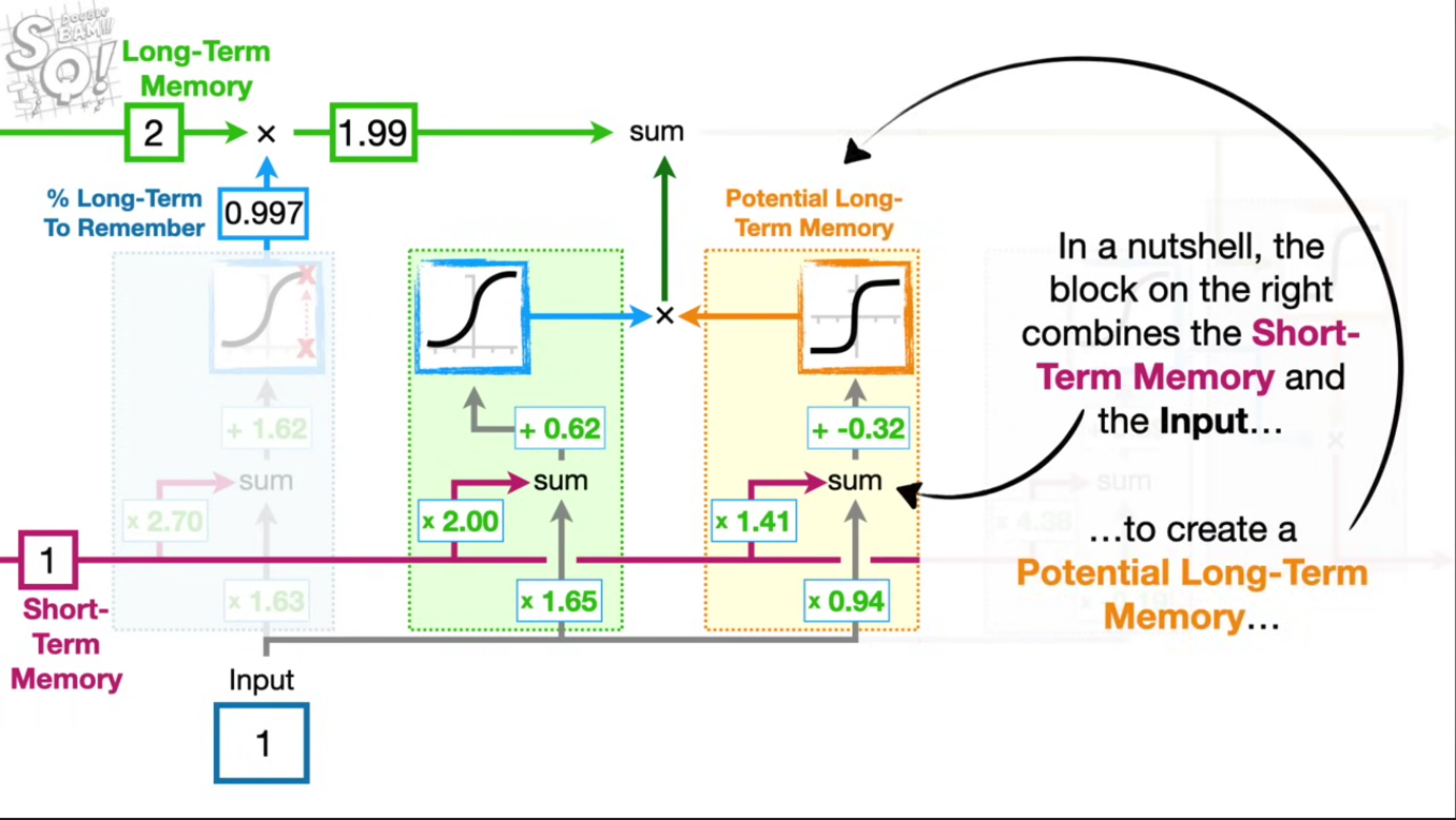


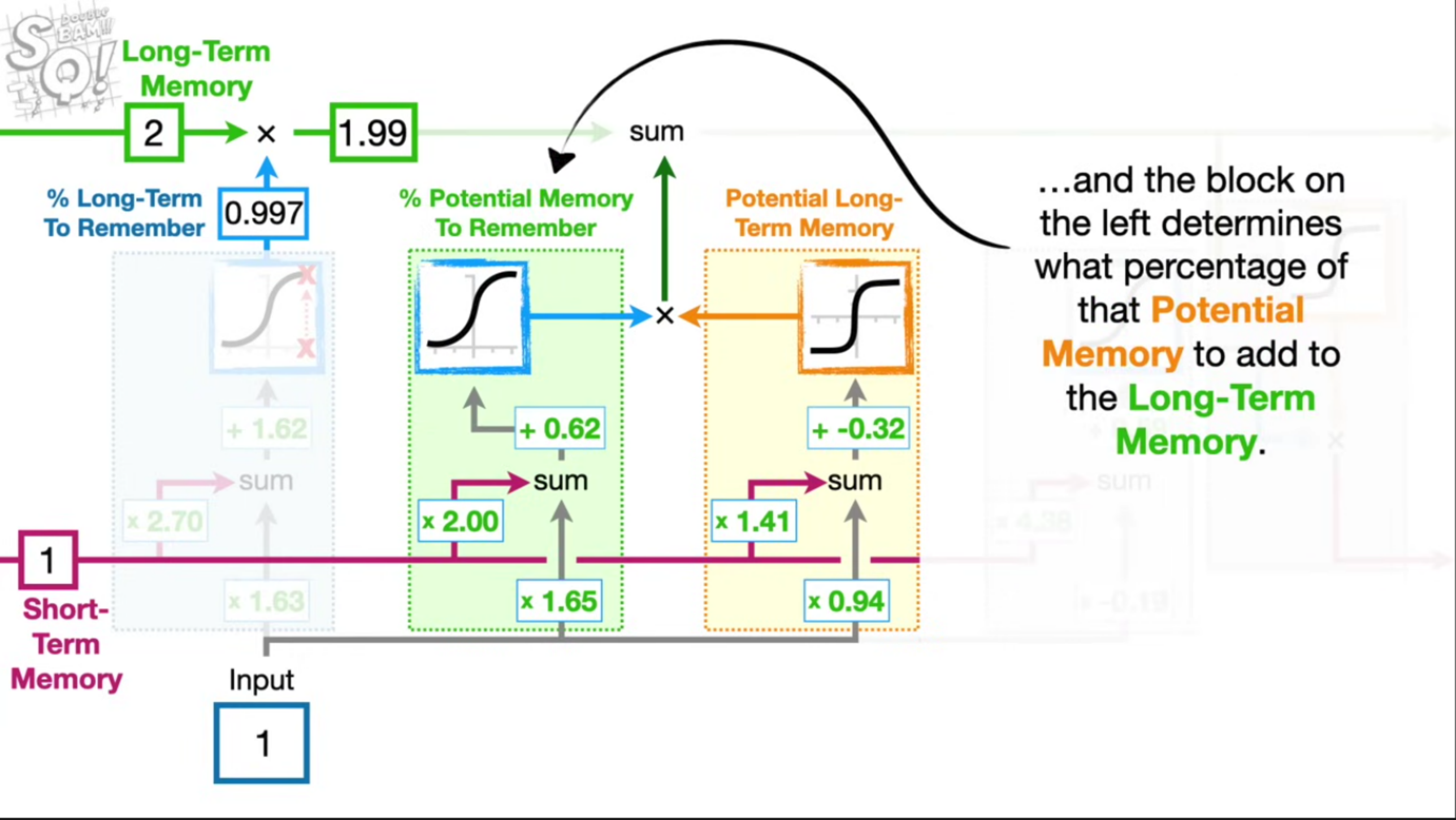
TERMINOLOGY ALERT!!!

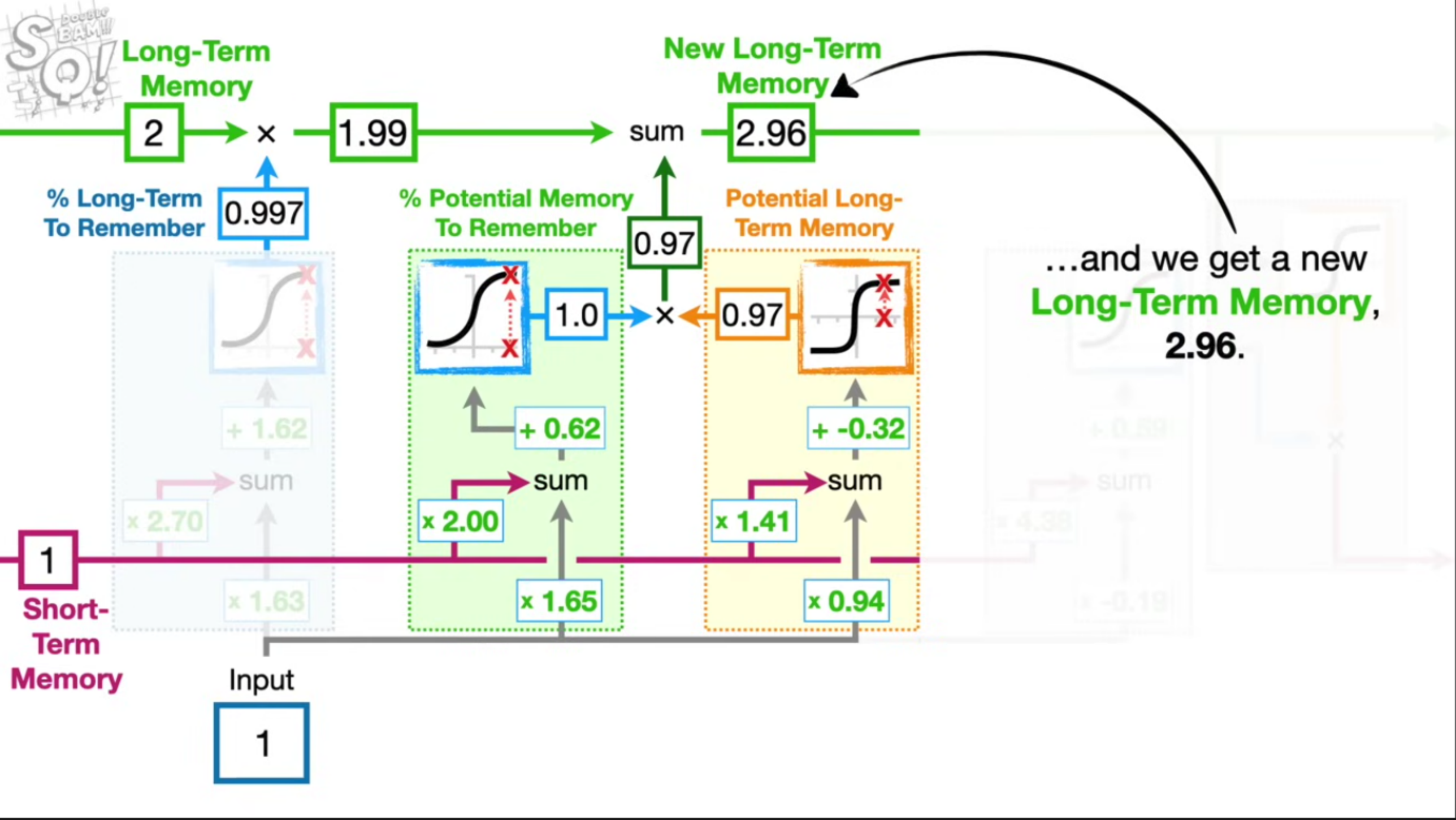
Even though this part of the **Long Short-Term Memory** unit determines what percentage of the **Long-Term Memory** will be remembered…

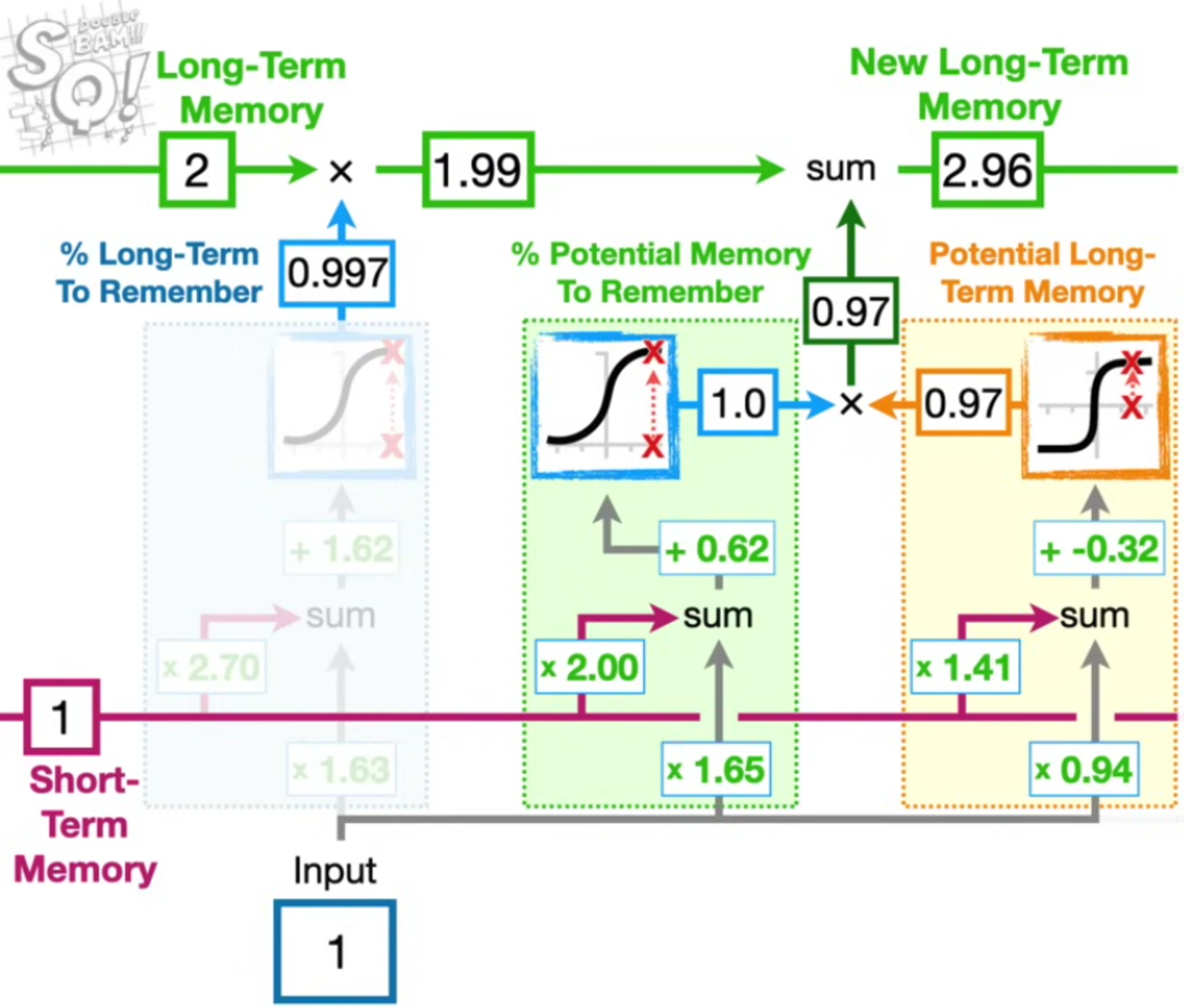
…it is usually called the **Forget Gate**.


Long-Term Memory
2
x
1.99
sum
% Long-Term To Remember
0.997
x
x
+ 0.62
+ -0.32
sum
sum
x 2.00
x 1.41
1
x 1.65
x 0.94
Short-Term Memory
Input
1
…and talk about what the second stage does.


Long-Term Memory
2
x
1.99
sum
% Long-Term To Remember
0.997
Potential Long-Term Memory
In a nutshell, the block on the right combines the Short-Term Memory and the Input…
+ 0.62
+ -0.32
sum
sum
x 2.00
x 1.41
1
x 1.65
x 0.94
Short-Term Memory
Input
1
…to create a Potential Long-Term Memory…


Long-Term Memory
2
x
1.99
sum
% Long-Term To Remember
0.997
% Potential Memory To Remember
Potential Long-Term Memory
+ 1.62
sum
x 2.70
x 1.63
+ 0.62
sum
x 2.00
x 1.65
+ -0.32
sum
x 1.41
x 0.94
1
Short-Term Memory
Input
1
…and the block on the left determines what percentage of that Potential Memory to add to the Long-Term Memory.

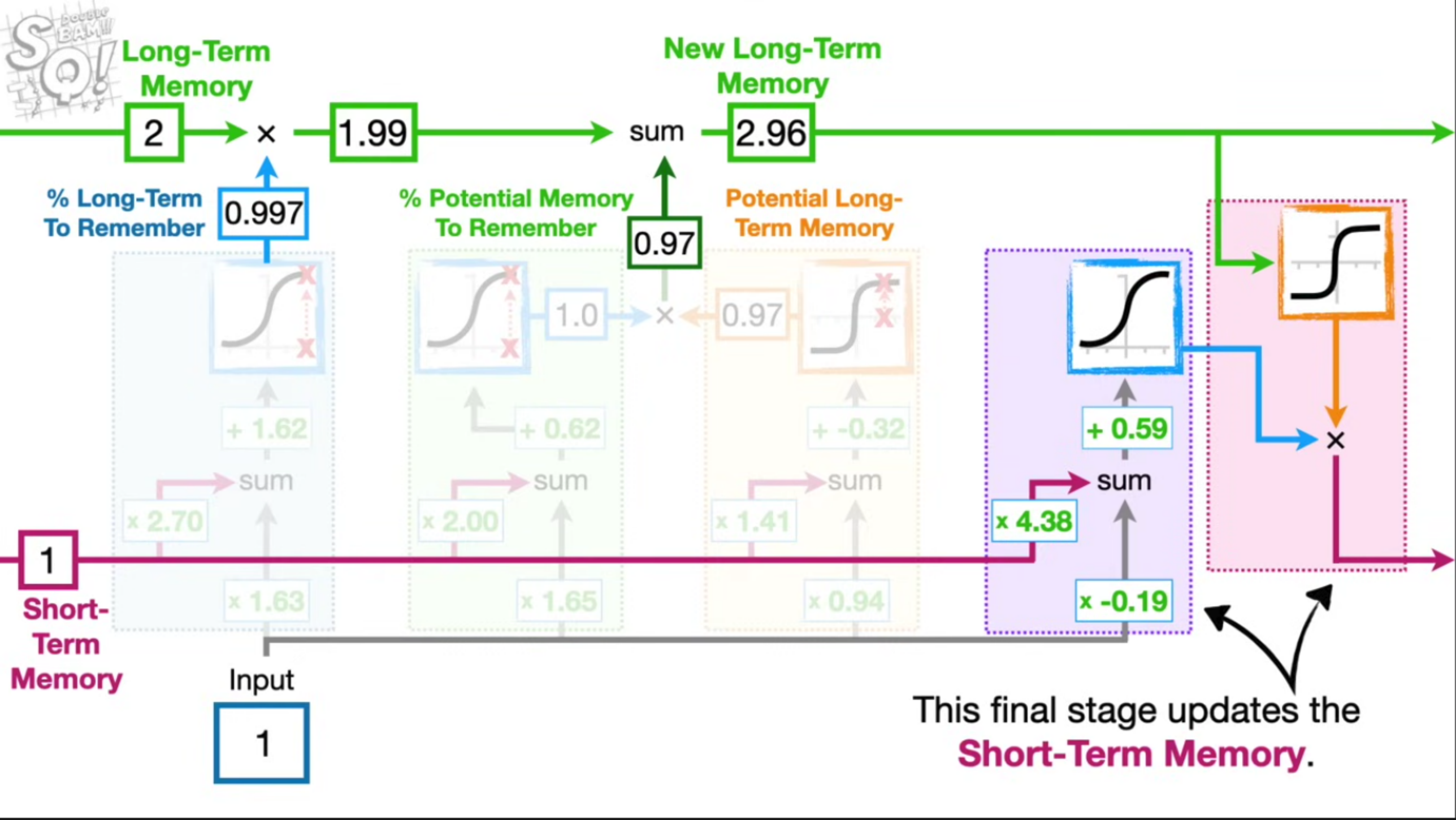

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
1.0
×
0.97
+ 0.62
+ -0.32
sum
sum
x 2.00
x 1.41
1
Short-Term Memory
x 1.65
x 0.94
Input
1
…and we get a new Long-Term Memory, 2.96.

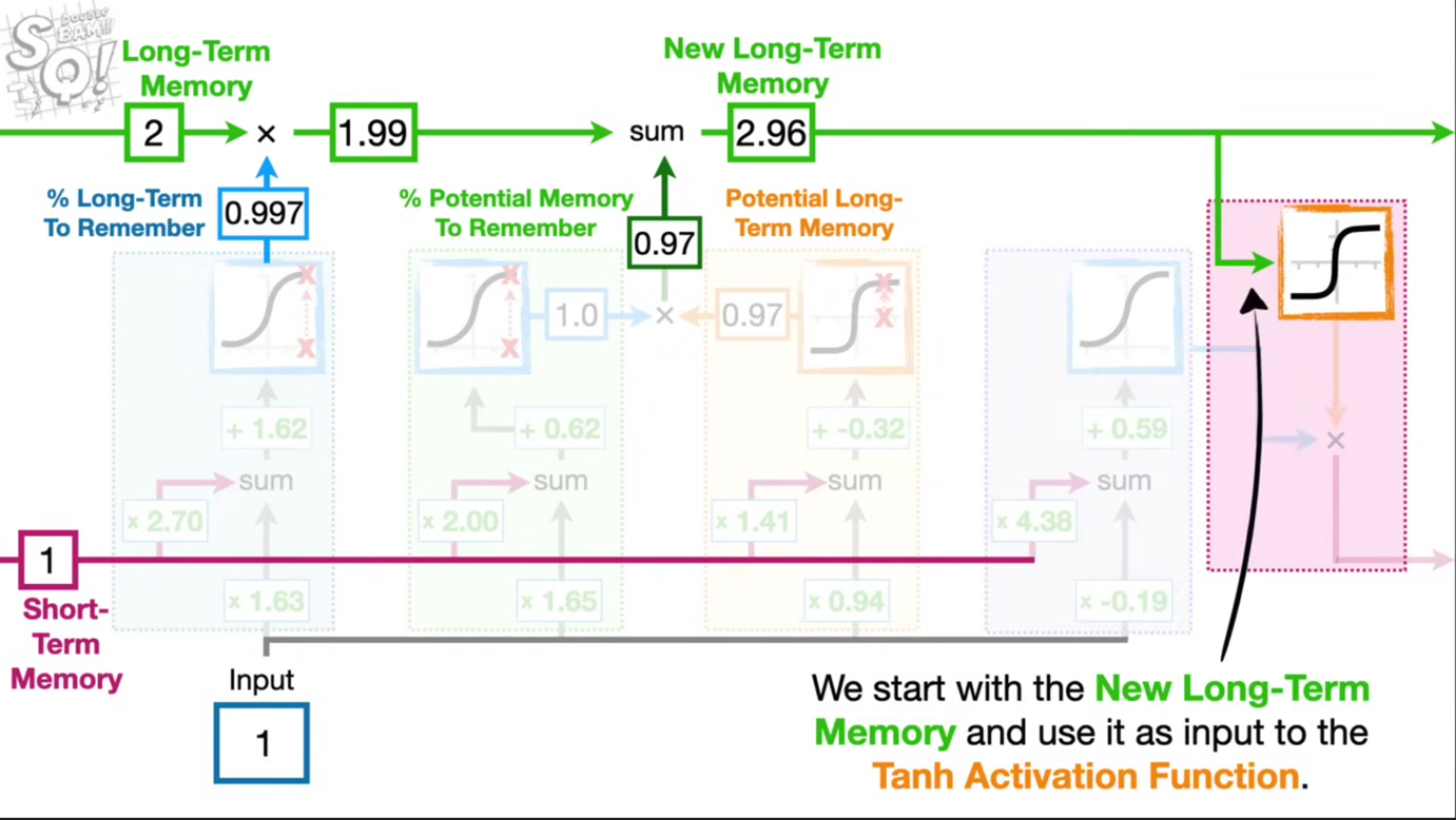


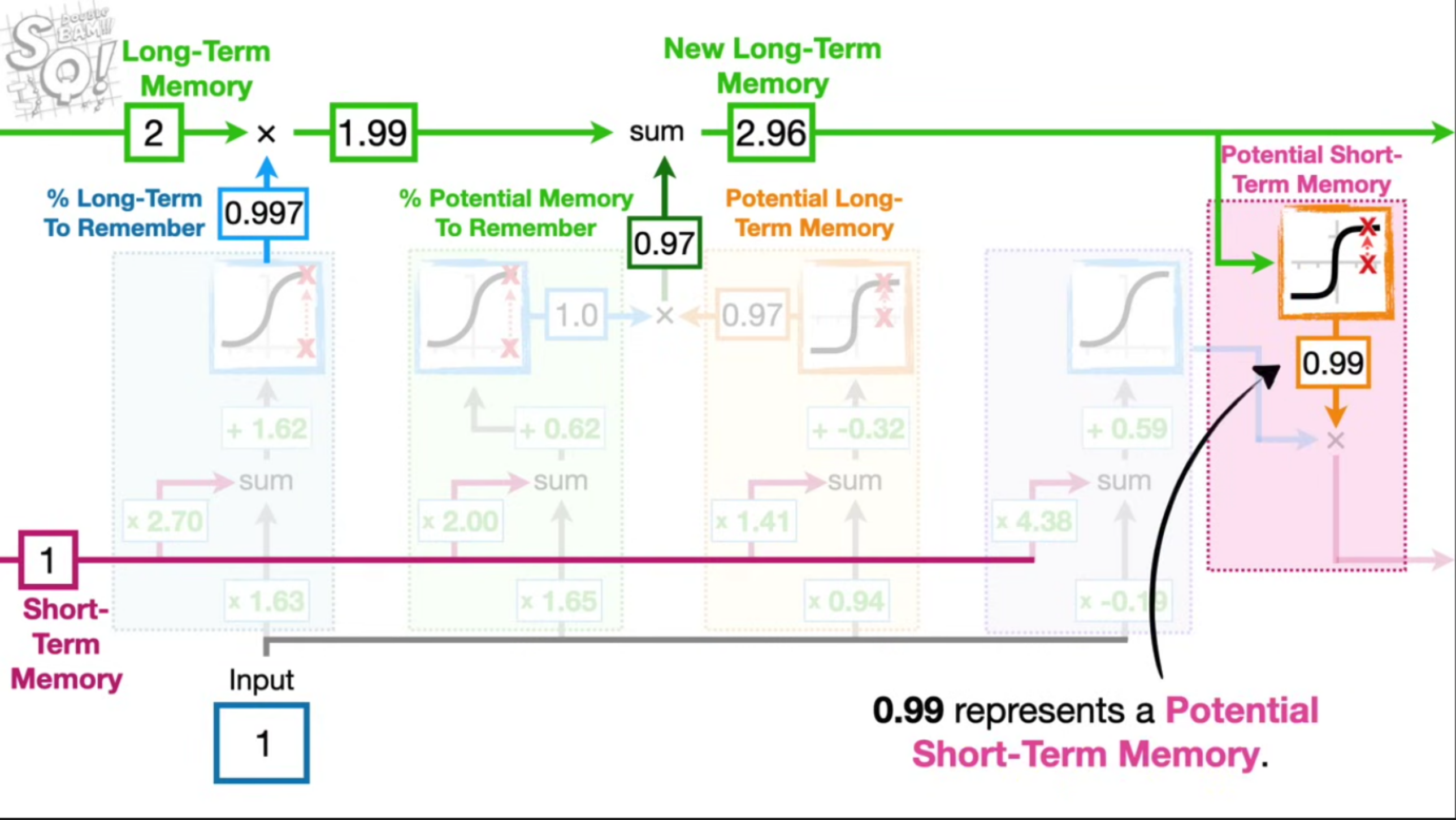
TERMINOLOGY ALERT!!!

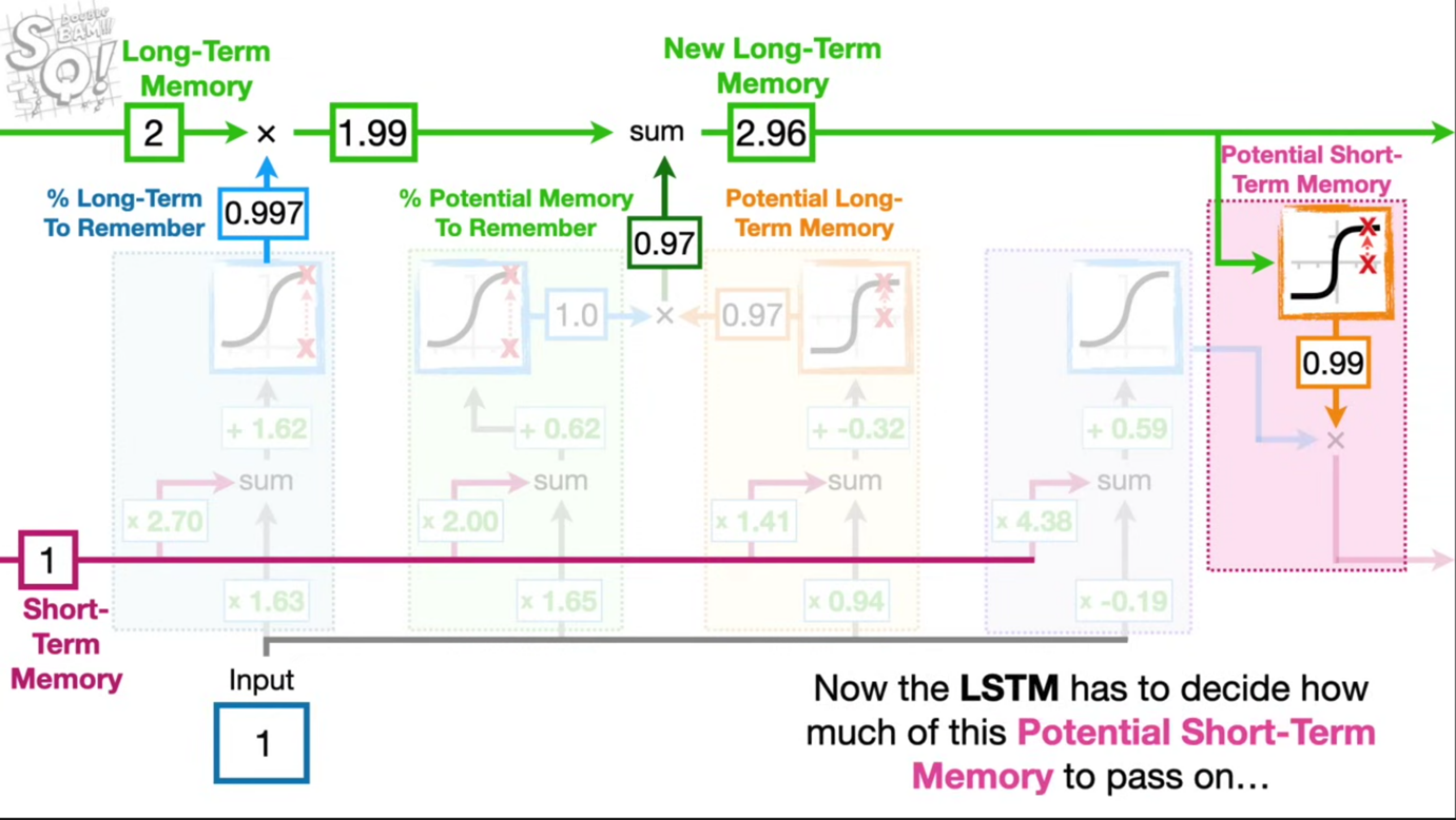
Even though this part of the **Long Short-Term Memory** unit determines how we should update the **Long-Term Memory**…

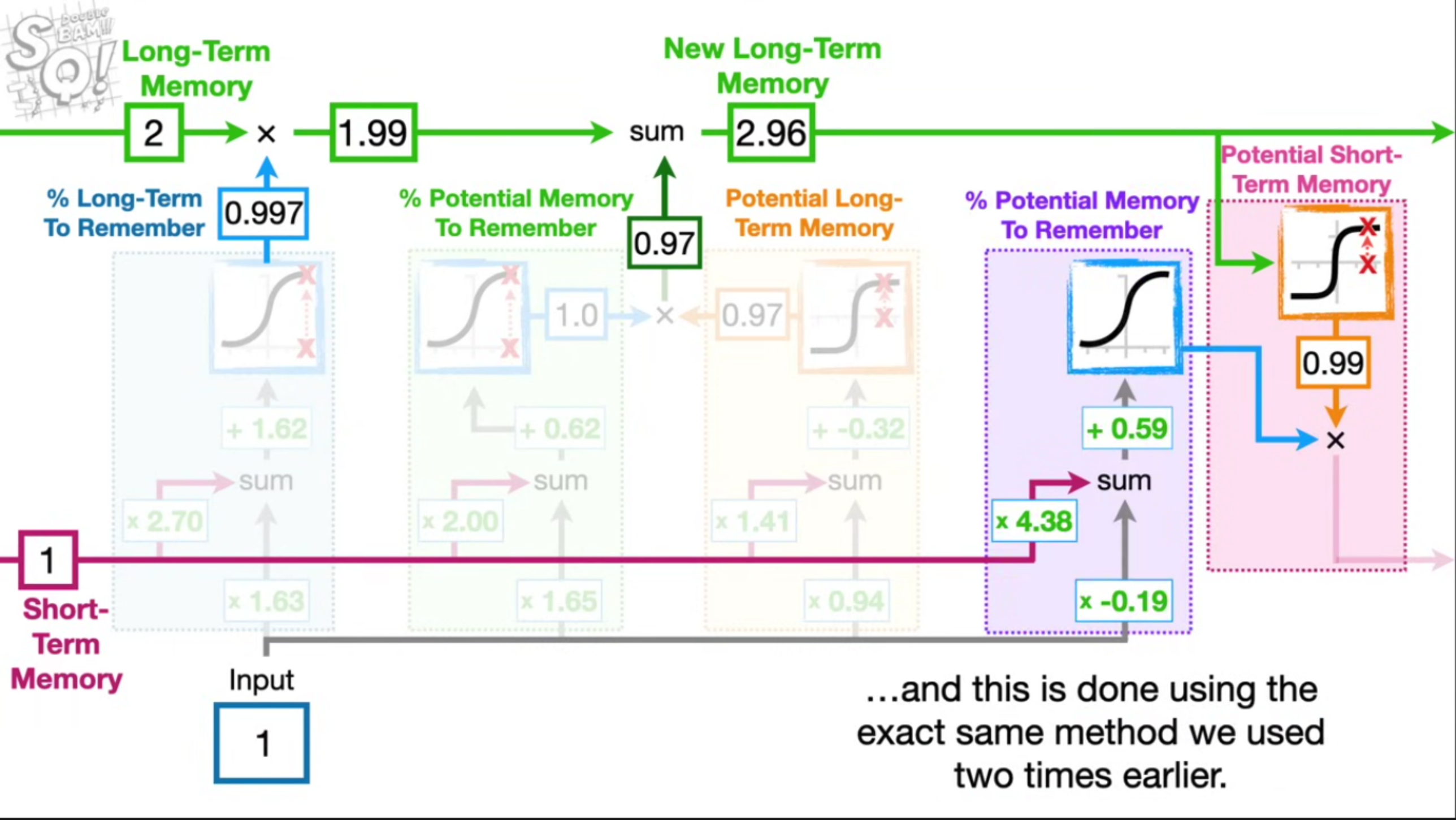
…it is usually called the **Input Gate**.

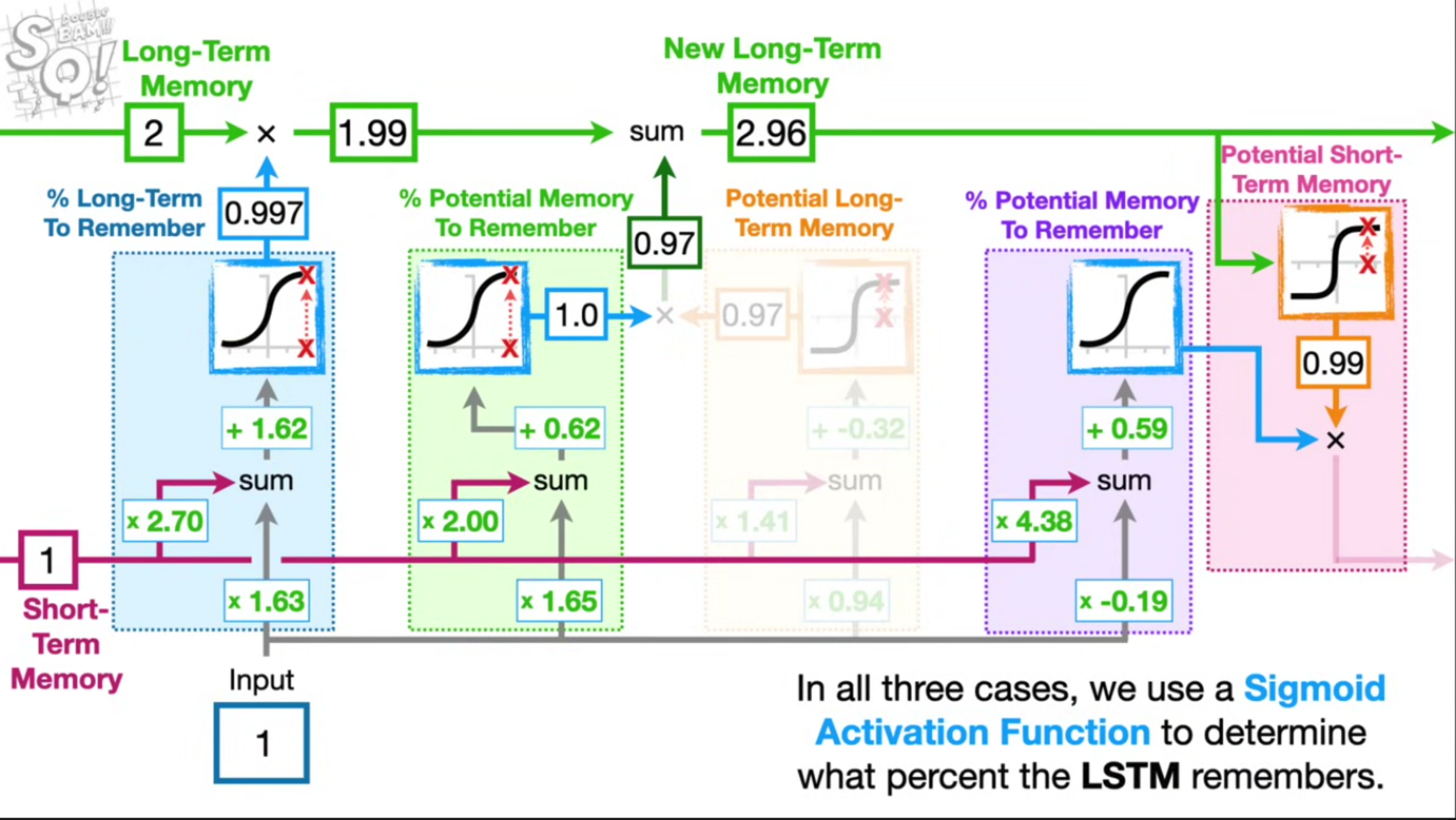

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
1.0
×
0.97
+ 1.62
sum
x 2.70
x 1.63
+ 0.62
sum
x 2.00
x 1.65
+ -0.32
sum
x 1.41
x 0.94
+ 0.59
sum
x 4.38
x -0.19
×
1
Short-Term Memory
Input
1
This final stage updates the Short-Term Memory.

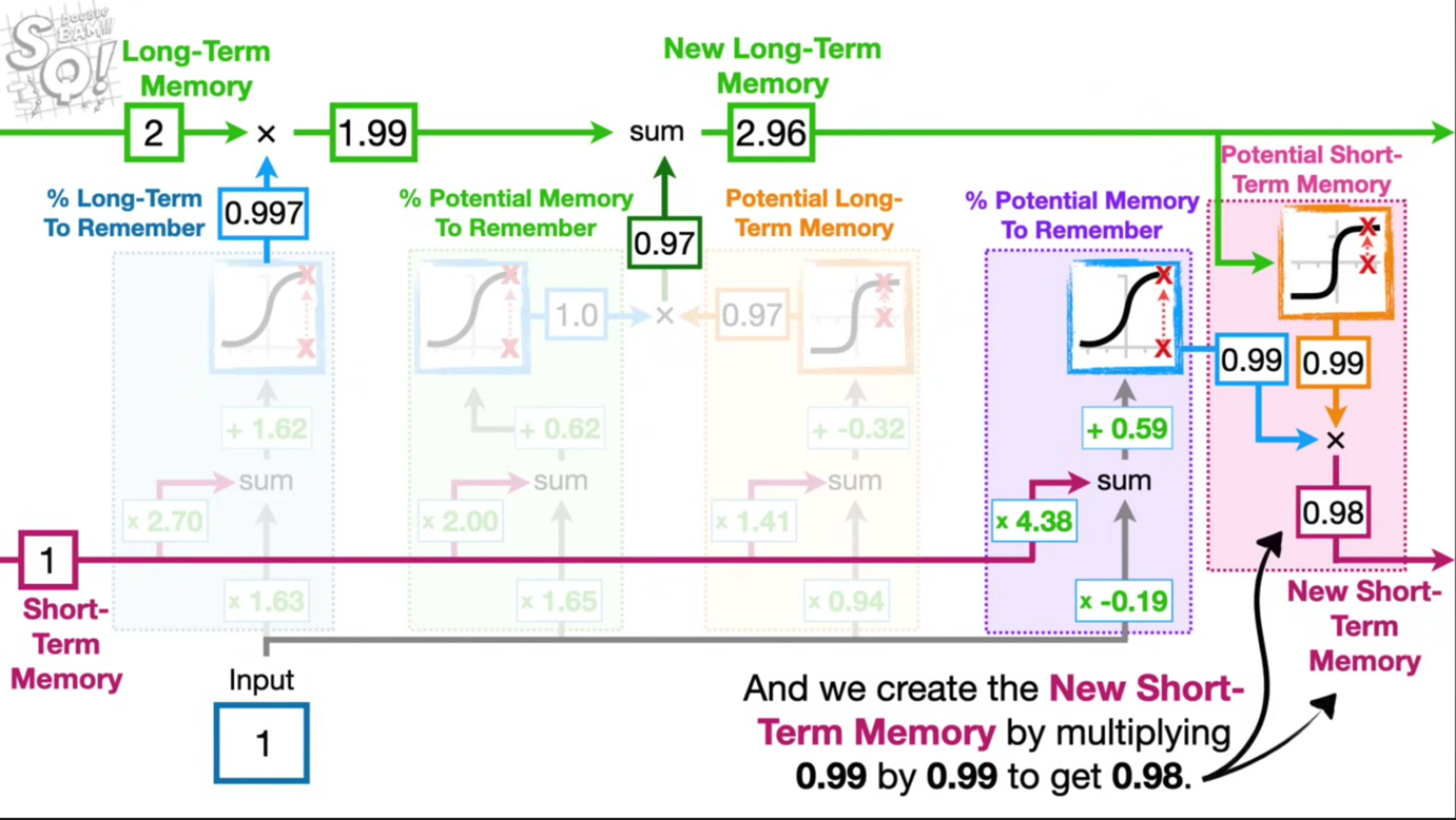

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
1.0
×
0.97
+ 1.62
sum
x 2.70
x 1.63
+ 0.62
sum
x 2.00
x 1.65
+ -0.32
sum
x 1.41
x 0.94
+ 0.59
sum
x 4.38
x -0.19
1
Short-Term Memory
Input
1
We start with the New Long-Term Memory and use it as input to the Tanh Activation Function.

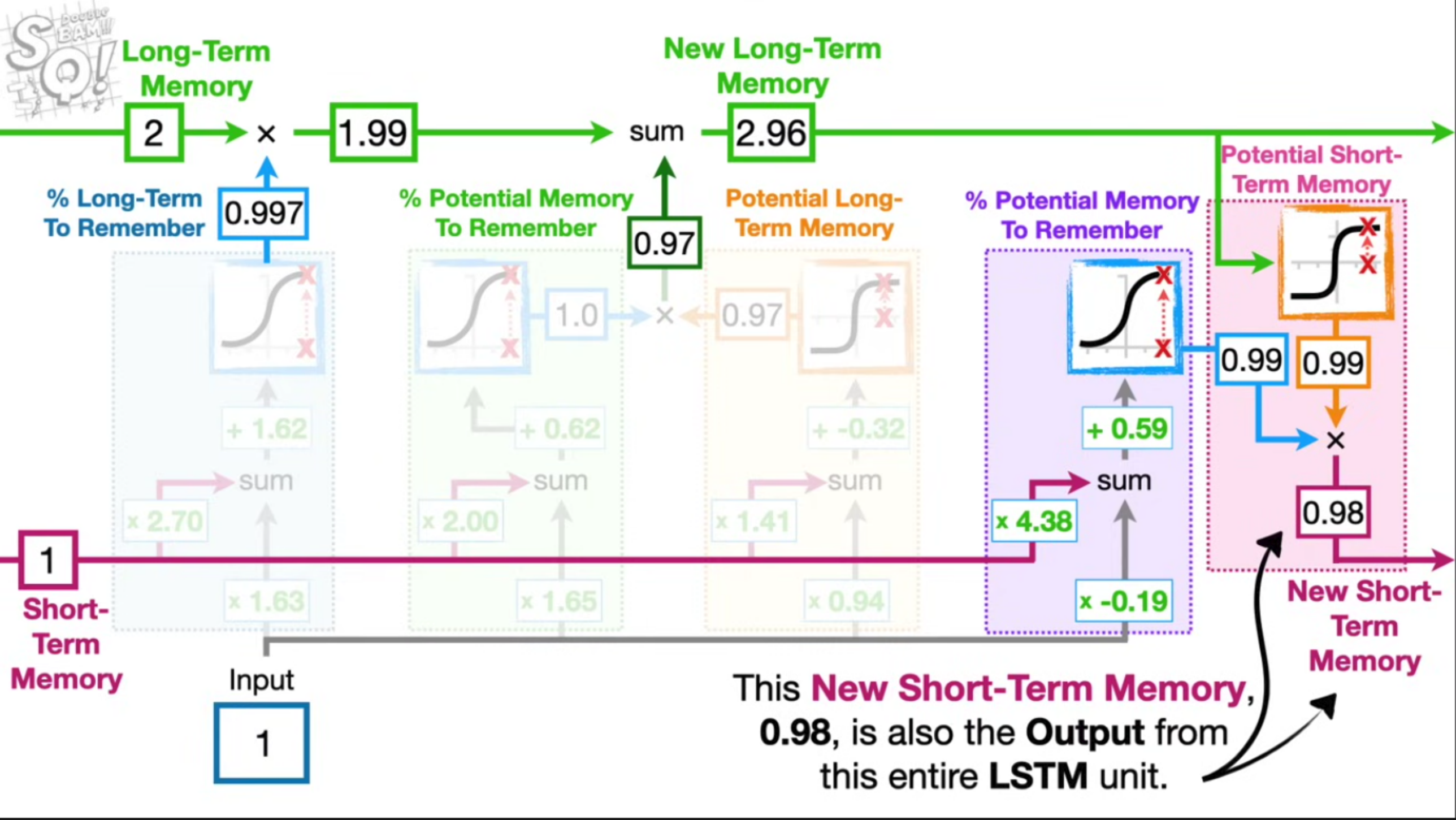

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
Potential Short-Term Memory
1.0
0.97
0.99
+ 1.62
+ 0.62
+ -0.32
+ 0.59
sum
x 2.70
x 2.00
x 1.41
x 4.38
1
x 1.63
x 1.65
x 0.94
Short-Term Memory
Input
1
0.99 represents a Potential Short-Term Memory.

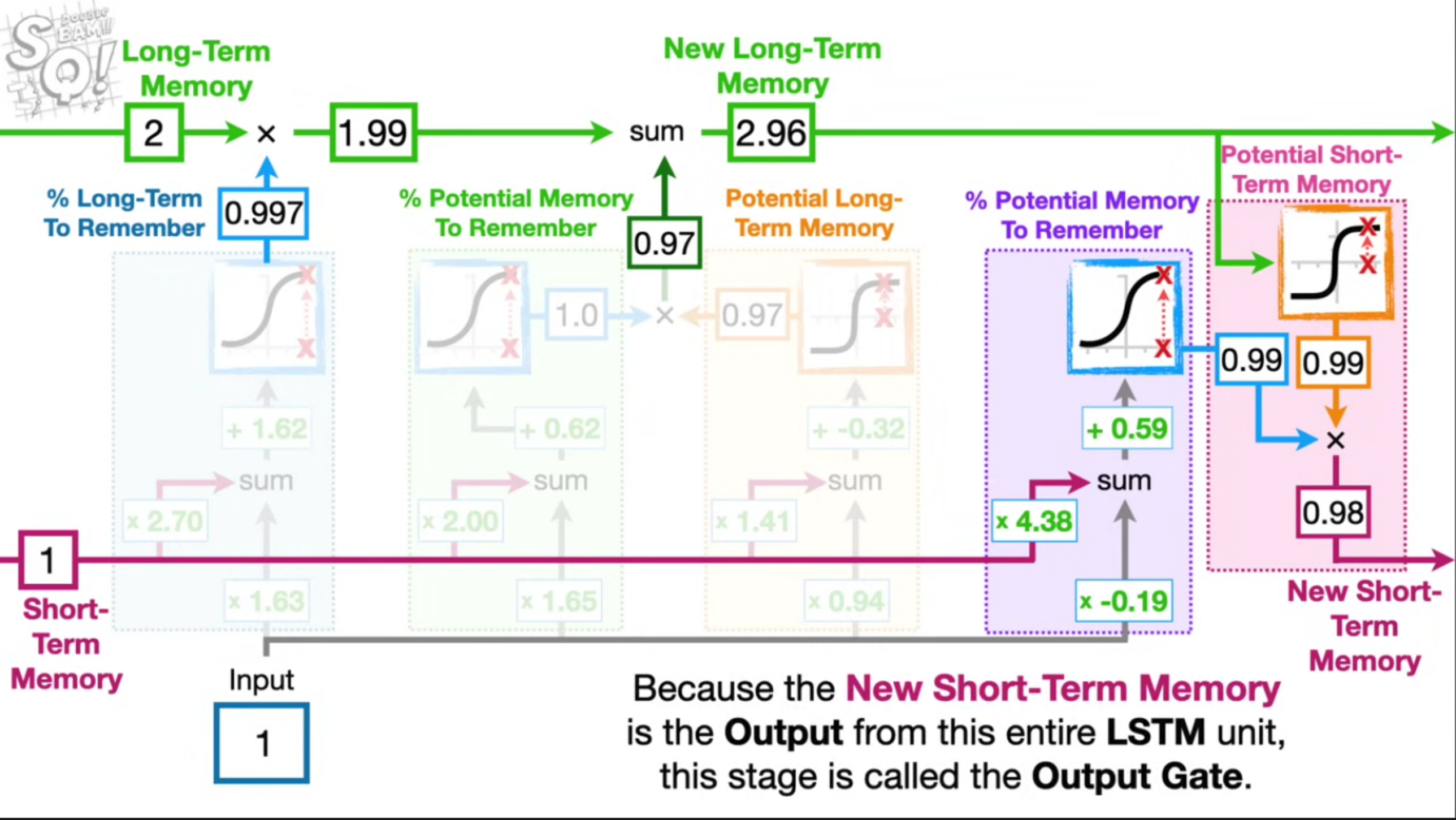

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
Potential Short-Term Memory
1.0
0.97
0.99
+ 1.62
+ 0.62
+ -0.32
+ 0.59
sum
x 2.70
x 2.00
x 1.41
x 4.38
1
x 1.63
x 1.65
x 0.94
x -0.19
Short-Term Memory
Input
1
Now the LSTM has to decide how much of this Potential Short-Term Memory to pass on…

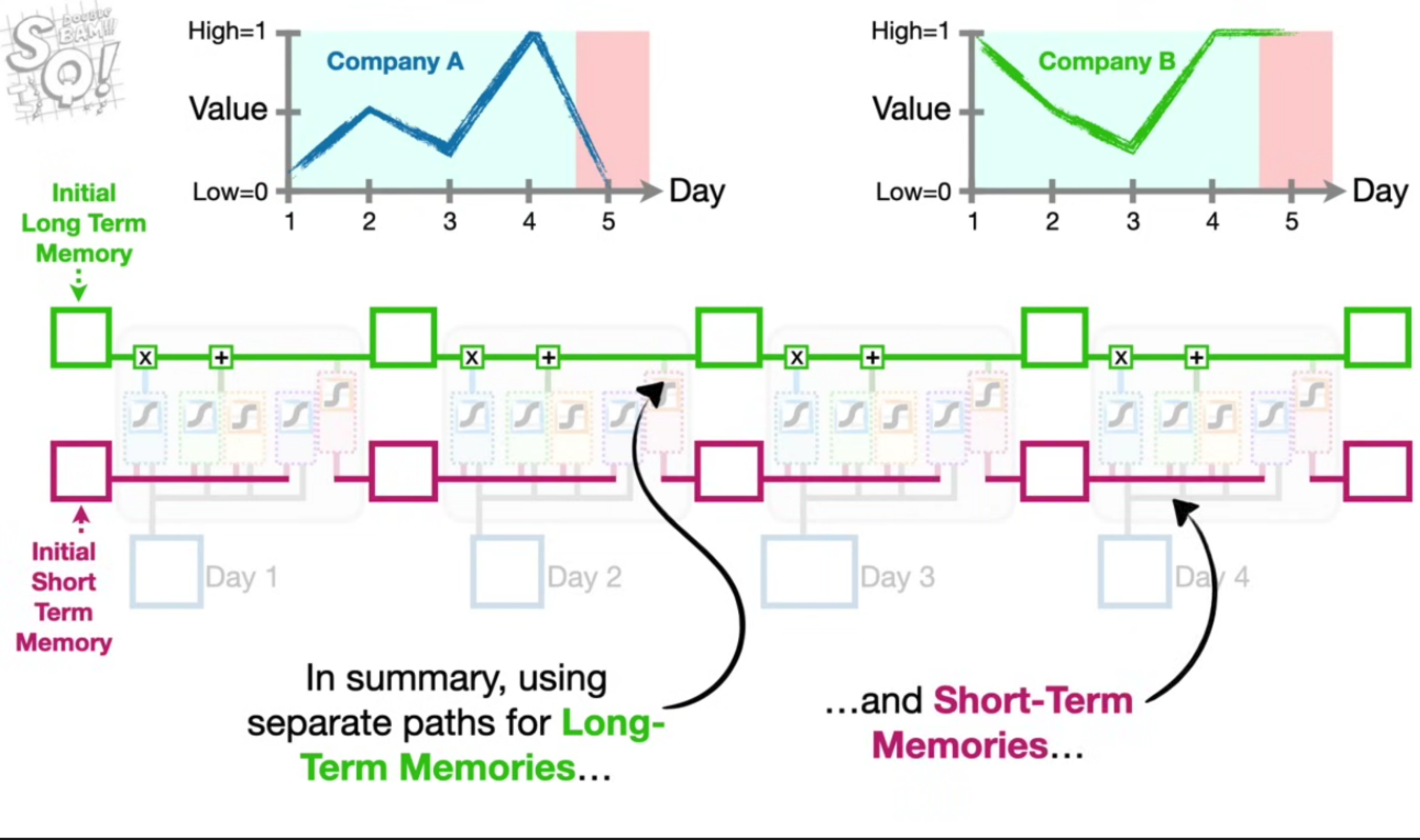

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
% Potential Memory To Remember
Potential Short-Term Memory
1.0
×
0.97
+ 1.62
+ 0.62
+ -0.32
+ 0.59
sum
sum
sum
x 2.70
x 2.00
x 1.41
x 4.38
x 1.63
x 1.65
x 0.94
x -0.19
0.99
×
1
Short-Term Memory
Input
1
…and this is done using the exact same method we used two times earlier.


Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
+ 1.62
sum
x 2.70
x 1.63
% Potential Memory To Remember
1.0
+ 0.62
sum
x 2.00
x 1.65
0.97
Potential Long-Term Memory
0.97
+ -0.32
sum
x 1.41
x 0.94
% Potential Memory To Remember
+ 0.59
sum
x 4.38
x -0.19
Potential Short-Term Memory
0.99
×
1
Short-Term Memory
Input
1
In all three cases, we use a Sigmoid Activation Function to determine what percent the LSTM remembers.


Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
% Potential Memory To Remember
Potential Short-Term Memory
+ 1.62
sum
x 2.70
x 1.63
1.0
×
+ 0.62
sum
x 2.00
x 1.65
0.97
+ -0.32
sum
x 1.41
x 0.94
+ 0.59
sum
x 4.38
x -0.19
0.99
0.99
×
0.98
1
Short-Term Memory
Input
1
New Short-Term Memory
And we create the New Short-Term Memory by multiplying 0.99 by 0.99 to get 0.98.

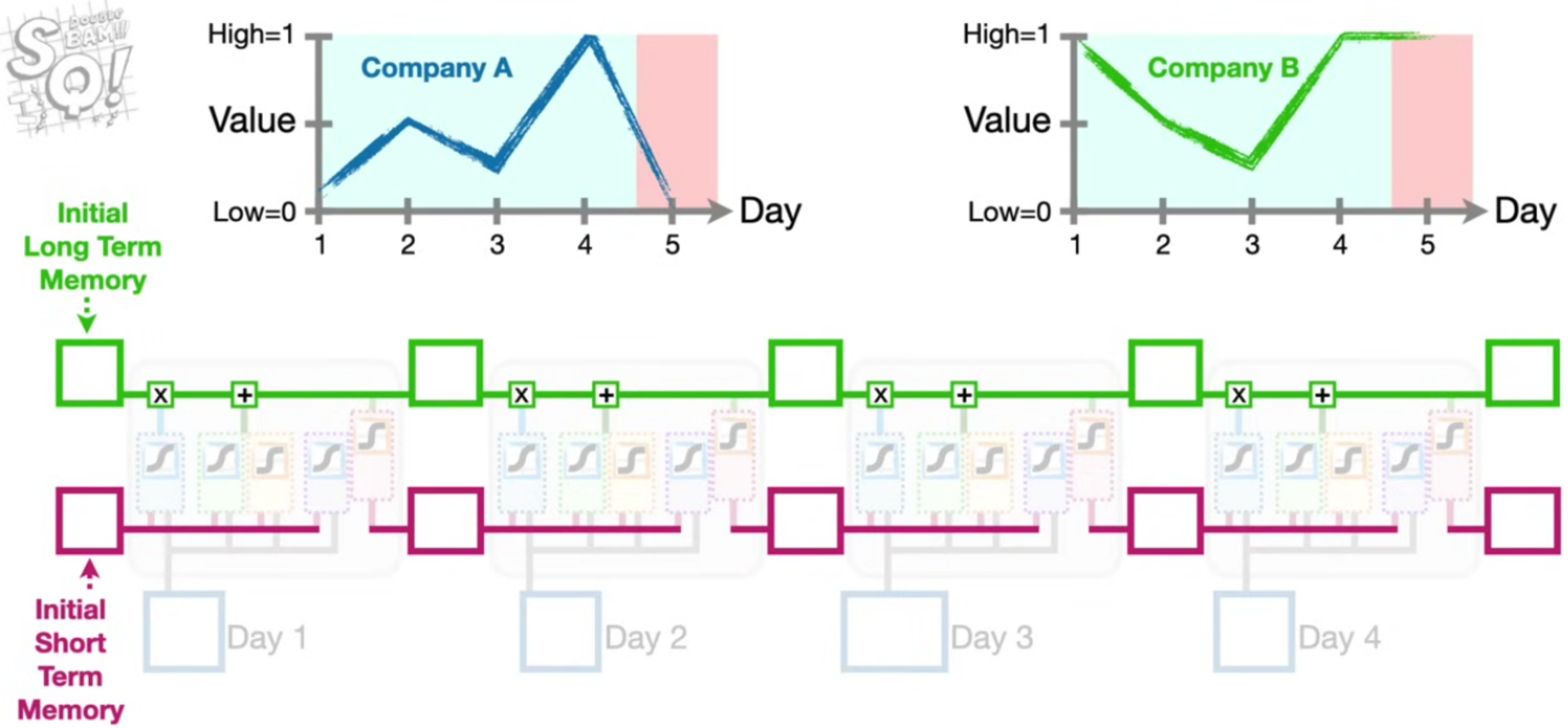

Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
% Potential Memory To Remember
Potential Short-Term Memory
+ 1.62
sum
x 2.70
x 1.63
1.0
×
+ 0.62
sum
x 2.00
x 1.65
0.97
+ -0.32
sum
x 1.41
x 0.94
+ 0.59
sum
x 4.38
x -0.19
0.99
0.99
×
0.98
1
Short-Term Memory
Input
1
New Short-Term Memory
This New Short-Term Memory, 0.98, is also the Output from this entire LSTM unit.


Long-Term Memory
2
×
1.99
sum
New Long-Term Memory
2.96
% Long-Term To Remember
0.997
% Potential Memory To Remember
0.97
Potential Long-Term Memory
% Potential Memory To Remember
Potential Short-Term Memory
+ 1.62
sum
x 2.70
x 1.63
1.0
×
+ 0.62
sum
x 2.00
x 1.65
0.97
+ -0.32
sum
x 1.41
x 0.94
+ 0.59
sum
x 4.38
x -0.19
0.99
0.99
×
0.98
1
Short-Term Memory
Input
1
New Short-Term Memory
Because the New Short-Term Memory is the Output from this entire LSTM unit, this stage is called the Output Gate.


High=1
Value
Low=0
Company A
1
2
3
4
5
Day
High=1
Value
Low=0
Company B
1
2
3
4
5
Day
Initial Long Term Memory
Initial Short Term Memory
Day 1
Day 2
Day 3
Day 4
In summary, using separate paths for Long-Term Memories…
…and Short-Term Memories…



…Long Short-Term Memory Networks avoid the **exploding/ vanishing** gradient problem…



…and that means we can unroll them more times to accommodate longer sequences of input data than a vanilla **Recurrent Neural Network**.